

Lunar construction: A state-of-the-art survey

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ABSTRACT

The renewed global interest in furthering human's presence on the Moon has catalyzed efforts to establish a sustainable lunar base. The incentive is not only for scientific opportunities and prospects of deep-space exploration, but also for demonstrating technologies that will extend our reach throughout the Solar System. Central to such efforts is the development of robust and scalable lunar construction technologies. This survey presents a comprehensive review of the state-of-the-art in lunar construction, including environmental characterization, infrastructure development, construction methods and materials, and robotic systems. The unique challenges posed by the lunar environment are highlighted, such as extreme temperature variations, high radiation exposure, and micrometeorite impacts, with a particular emphasis on the abrasive, adhesive, and electrostatically charged lunar regolith, thus including strategies developed for lunar dust mitigation. The survey investigates the critical infrastructure that will need to be established, including habitats, power stations, communication stations, landing pads, blast berms, and more. A detailed analysis of the methods and materials that are being developed to create such infrastructure is presented, identifying which methods have demonstrated promise and garnered the most attention. A diversity of robotic technologies are required to enable the construction of the necessary infrastructure using these methods and systems, which are broken down into lunar cranes, mobile manipulators, 3D printers, and robot teams, with a particular focus on work being done to develop flight systems. The paper concludes by identifying critical research and technological gaps that must be addressed to support the next generation of lunar missions and long-term extraterrestrial habitation.

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1. Introduction

The recent renewal of focus on the Moon by space agencies and private sector is motivated by not only its scientific opportunities and prospects of deep-space exploration, but also demonstrating technologies that will extend sustained human presence throughout the Solar System. The International Space Exploration Coordination Group (ISECG), a consortium consisting of 27 space agencies, revised its Global Exploration Roadmap (GER) in 2024 with a supplement updating mission scenarios and architecture for both human and robotic lunar surface missions [1]. The goal is to ensure a sustainable return to the Moon by establishing the required infrastructure, and to highlight the increasing interest in certain technologies such as lunar in-situ resource utilization (ISRU), while generating socio-economic benefits on Earth. Some notable efforts to pave the roadmap include the NASA-led Artemis Program, whose one of the key goals is to establish Sustained Artemis Base Camp Infrastructure and Commercial Operations at the lunar South Pole by the late 2020s, as a starting point for the future Mars-based infrastructure [2]. In a similar concept, the European Space Agency (ESA) has proposed the establishment of the Moon Village habitat in conjunction with the Massachusetts Institute of Technology (MIT), though no specific target date has been yet released [3–6]. The Chinese National Space Administration (CNSA) and Russia's Roscosmos [7], in collaboration with 11 other nations to date [8], have also expressed their goal to send humans to the Moon before 2030, and to establish an International Lunar Research Station on the lunar surface. India, a relatively new player in lunar exploration, has also cited the long-term goal of establishing a lunar base in the Indian Space Research Organization (ISRO) Vision 2047 [9]. As part of the renewed GER, Phase 1 of the lunar surface exploration and utilization, i.e., Boots on the Moon, heavily relies on a near-term series of robotic missions for scientific and technology demonstration objectives as well as creating the basic elements of lunar infrastructure. The Canadian Space Agency (CSA) has made its commitment to the development of lunar infrastructure through the Lunar Surface Exploration Initiative (LSEI), where Robotics is presented as one of the key subjects of study [10].

A major challenge in establishing the lunar infrastructure is the scale of the work required for lunar construction. Planetary missions for years has been focused on exploration, where the primary goal is merely scientific, and machines and equipment need to minimally manipulate and interact with local resources. The construction of lunar infrastructure, on the other hand, will require extensive in-situ resource utilization (ISRU) reliably and at scale. This necessity brings to light our limited knowledge of and experience with the lunar environment, construction, and robotics. Some reviews already exist, such as [11] and [12], which focus on lunar habitat concepts and lunar construction methods and materials. Other reviews, such as [13] and [14] focus specifically on the development of lunar construction methods and

materials, with the latter focusing on NASA's Moon-to-Mars Planetary Autonomous Construction Technology (MMPACT) development. Another review [15] focuses on the development of lunar crane technologies. Nevertheless, little work has been done to gather the global state-of-the-art work in lunar construction and related technologies into one comprehensive survey.

There are several necessary components of lunar construction for establishing a lunar habitat, as depicted in Fig. 1. First, there is a need for the survey of the lunar surface where habitats are likely to be constructed (Section 2), including characterization of the lunar environment (Section 2.1) as well as lunar regolith (Section 2.2), which greatly impacts the surface systems. The impact is significant to the extent that it requires dedicated technologies to mitigate lunar regolith dust (Section 2.3). Secondly, lunar infrastructure will need to be established on the lunar surface, considering the favorable sites based on terrain, power availability and other factors (Section 3). The required infrastructure includes habitats (Section 3.1), power stations (Section 3.2), communication stations (Section 3.3), landing pads (Section 3.4), blast shields, and roads (Section 3.5). Thirdly, lunar construction methods and materials (Section 4) should be developed for building the surface structures, including various lunar cements (Section 4.1), additive manufacturing technologies (Section 4.2), and the utilization of raw lunar regolith (Section 4.3). This survey also includes an analysis of the work that has been done to advance these construction methods and materials in recent years (Section 4.4). Finally, construction systems (Section 5) are required to conduct most of the construction activities, especially during the preparatory phases, such as regolith excavators (Section 5.1), conveyors (Section 5.2), and various types of construction robots (Section 5.3). It is worth noting that while this review discusses the methods and materials for constructing lunar surface infrastructure, it excludes the discussion on the sourcing of ISRU materials.

2. Lunar site surveying

In general, most international efforts have focused on the lunar South Pole as the primary location for a sustainable lunar habitat. This is due to a variety of factors, including illumination and power profiles, the presence of volatiles (e.g., water ice) in Permanently Shadowed Regions (PSRs), and terrain [4,16–20]. Some works have studied the illumination conditions at both polar regions and how these can impact power generation [19,20]. The illumination conditions at the lunar poles are severely impacted by the local terrain, owing to the low inclination angle of the Moon's rotational axis at 1.54° [21]. This inclination creates two phenomena of interest at the lunar poles. The first phenomenon is *Peaks of Eternal Light* or *Peaks Of Light* (POL), where the terrain elevation is higher than the horizon, allowing for permanent sunlight and solar power generation. The second phenomenon involves PSRs, where there are permanent shadows in the craters, which can trap essential resources and volatiles such as water ice. An ideal location would include a region of near permanent sunlight for power generation, next to a PSR for resource extraction, and relatively

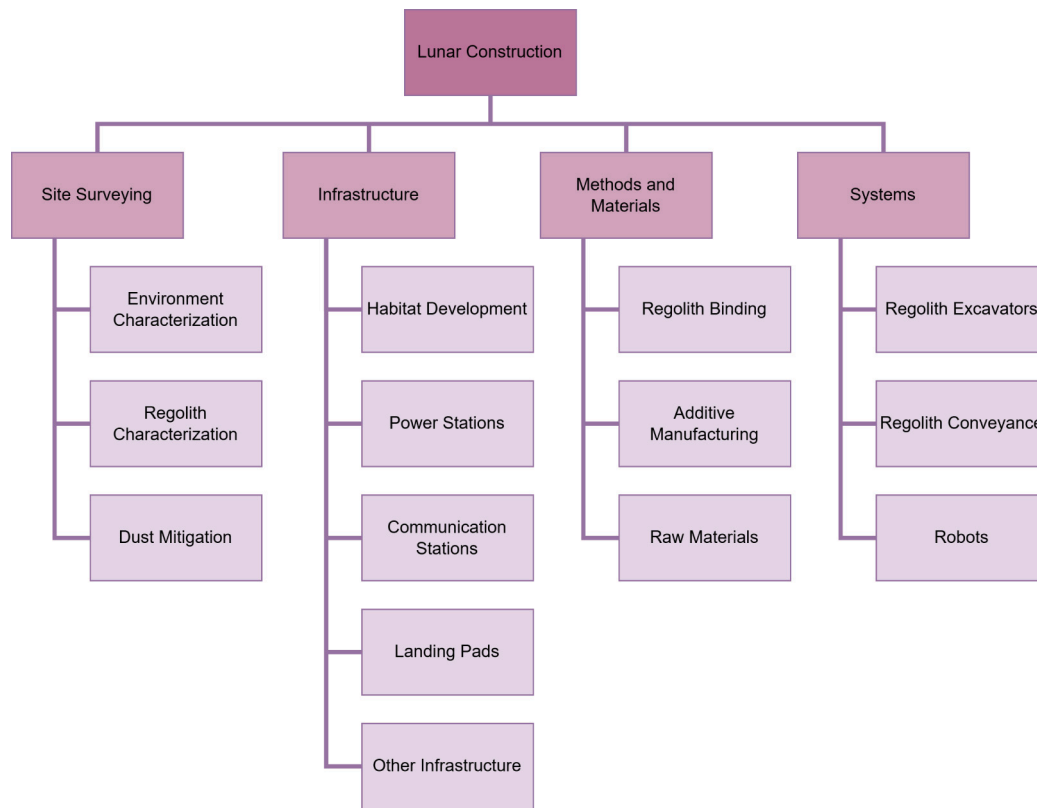


Fig. 1. Lunar Construction Overview.

flat terrain for habitat construction and landing [22,23]. Some works have also considered the implementation of solar towers, ranging from 2 m in height up to 2000 m, to induce permanent sunlight at the solar array locations, achieving constant or near-constant illumination in some cases [24,25]. However, these strategies can also lead to additional challenges, such as casting large shadows by solar towers across the region of interest, due to the low incidence angle of the Sun on the lunar horizon [25]. Small scale solar towers are already being designed, such as the Lunar Array, Mast, and Power System (LAMPS), which can extend up to 15 m height [26].

2.1. Lunar environment

Some overviews of the lunar environment are presented by [27] and [28]. The primary source of data stems from the “Lunar Sourcebook” [29], which curates and summarizes the findings of the Apollo missions and subsequent analysis of the lunar environment and regolith. The lunar gravity is approximately $1/6$ g, or 1.63 m/s^2 . The atmosphere is almost non-existent, and is largely composed of neon, hydrogen, helium, and argon, with a total mass of approximately 25000 kg and pressure of 3×10^{-15} atm.

The Moon has no electromagnetic field, so it is constantly bombarded by radiation from space. The radiation includes solar flares (called Solar Particle Events, SPEs), solar wind, galactic cosmic rays (GCRs), and radiation from the Moon [27,28]. Galactic cosmic rays and solar wind are inversely proportional, since solar wind deflects GCRs due to expanding the heliospheric magnetic shield. Solar wind and solar flares operate on an 11-year activity cycle. In general, humans have a maximum radiation allowance of 0.05 Sv/year, where 0.1 Sv results in a 0.5% increase in the chance of cancer later in life. The GCRs alone cause 0.3 Sv/year on the surface, and solar flares cause up to 0.6 Sv/h. This can be compared to the 0.001–0.002 Sv/year that the average human experiences on Earth. To mitigate the radiation effects, regolith shielding can be used. While the Earth’s atmosphere provides 1000

g/cm^2 of shielding, 2–3 m of regolith provides 40 g/cm^2 . (The unit is a common measure of radiation shielding, which is agnostic of depth that can be recovered using material density.) It is commonly accepted that 1 m of regolith shielding would be sufficient for nominal conditions, but rare large flares would require 700 g/cm^2 of shielding [30].

Temperatures were found to vary by 282°C on the lunar surface at the Apollo landing sites, from -171°C to 111°C [30], dropping below -213°C in PSRs near the poles [31]. This variation is drastically reduced with depth, making the lunar regolith a good thermal insulator. At the Apollo landing sites, where the equilibrium temperature of lunar regolith is in the range of -15°C to -20°C , the variation from the equilibrium is nearly 10°C at 20 cm depth, and 1°C at 40 cm depth [32]. A comprehensive mapping of lunar surface temperatures is compiled in [31], using 10 years of Diviner data from the Lunar Reconnaissance Orbiter, indicating that the temperature range and distribution remain similar between the North and South poles throughout seasons, with slight increases in the size of cold spots during the winter. This consistency can be explained by the Moon’s lack of weather and low axial tilt. Temperatures vary from over 27°C in the hottest regions to below -231°C in the coldest regions, the latter typically coinciding with PSRs in the craters. The maximum bolometric temperatures (based on the total energy emitted across all wavelengths) for the North and South poles are shown in Fig. 2, and the presence of lunar ice in Fig. 3, indicating a large overlap between the coldest regions and water ice, particularly at the South Pole. Operations in these regions under cryogenic conditions are particularly challenging, requiring the development of novel energy management technologies, robotic systems, and structural materials to address the needs of lunar infrastructure and habitats.

Seismic activities on the Moon are limited, including deep and shallow “moonquakes” as well as meteorite impacts. The Apollo missions collected readings of approximately 12000 moonquakes, with most measuring 1–2 on the Richter scale, and a maximum of 3. Seismic activities are dominated by the meteorite impacts experienced on the

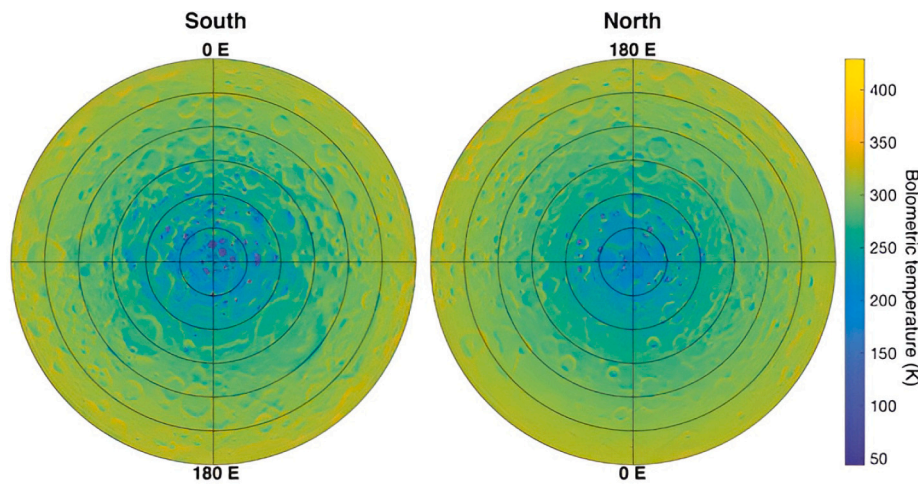


Fig. 2. Maximum Bolometric Polar Temperatures [31].

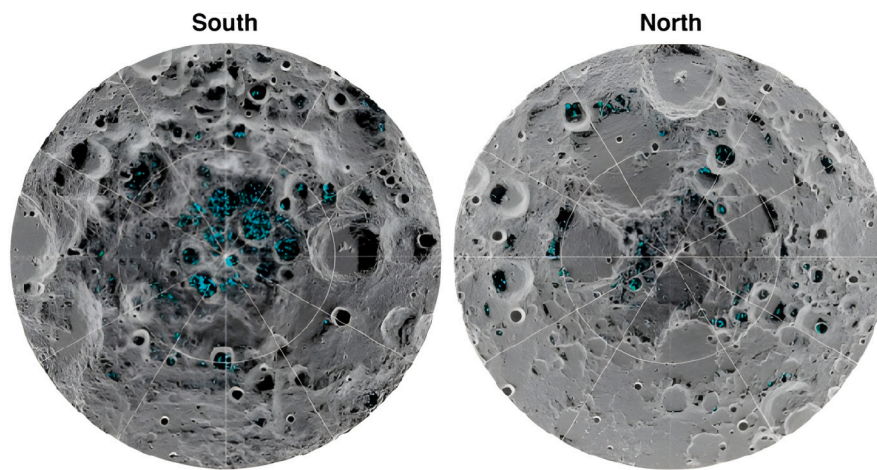


Fig. 3. Distribution of Polar Surface Ice [36].

surface [33]. Micrometeorites have an average mass of 10^{-10} to 10^{-8} kg, and an average speed of 72 km/s [27,34]. In general, structures can be protected from micrometeorites by lunar regolith and structural elements, such as Whipple shields [35].

The lunar crust is composed of 60% oxygen by mass. The oxygen is in the form of SiO_2 (40-50% of the lunar crust), FeO (10-20%) and/or CaO (10-20%). Lunar water ice is identified at the polar regions, particularly the South Pole, making it an attractive target for settlement [28]. NASA's Moon Mineralogy Mapper instrument aboard Chandrayaan-1 was able to collect data confirming the presence of lunar ice at the polar regions, shown in Fig. 3 [36].

2.2. Lunar regolith

The surface of the Moon has been pulverized by the collisions of both small and large asteroidal and cometary materials for billions of years, forming a fine-grained layer of "regolith" [29]. As the high-speed meteorites and micrometeorites impacted the surface, the soil particles broke up and began melting together, forming lithic fragments and clustering to form larger petrological elements, such as glass and agglutinates. The lunar regolith has a consistent particle and petrological distribution with depth, but is locally poorly sorted [37]. Overall, lunar regolith petrology can be broken down into a few groups: lithic fragments, breccia fragments, mineral fragments, glass fragments, and agglutinates. The lunar regolith itself is a generally fine, adhesive, sticky, grainy, and highly abrasive material, which has

properties unlike typical terrestrial soil. Several recent reviews of lunar regolith and its properties, specifically relating to engineering design and excavation, have been conducted by [37], [38], [39], and [40].

With respect to the abundance of regolith on the surface of the Moon, one study estimates the regolith median depth to be 2-4 m in the maria and 6-8 m in the highlands [41]. A more recent study estimates an average rock-free depth of 10 m in the maria and 12 m in the highlands [42]. Particle size properties remain consistent up to several meters of depth [39]. While some studies have attempted to fit distributions to the various properties of lunar regolith, such as size distribution, these models are heavily dependent on the samples that have been collected and analyzed [43,44]. The work of [34] details many important properties of the lunar regolith and dust, such as its dynamics and mechanics, including dust charging, which can impact its movement. Generally, the particle size characterization of lunar regolith can be broken down into three categories: lunar dust ($< 20 \mu\text{m}$), lunar soil ($< 1 \text{ cm}$), small rocks ($< 1 \text{ m}$), and mega-regolith ($> 1 \text{ m}$) [45]. Lunar regolith particles are extremely fine, with an average size of $70 \mu\text{m}$ and average elongation of 1.35, making them ovular. The typical lunar regolith particle has nearly eight times the surface area as an equivalent sphere, contributing to its high interlocking behavior. Regolith particle interlocking results in packing that can make regolith properties anisotropic [38]. The regolith consists of a loosely packed upper layer, which rapidly compacts with depth past 15 cm. The bulk density (mass per total volume) stands at 1.3 g/cm^3 at the surface, 1.52 g/cm^3 at 10 cm depth, and 1.83 g/cm^3 at 100 cm.

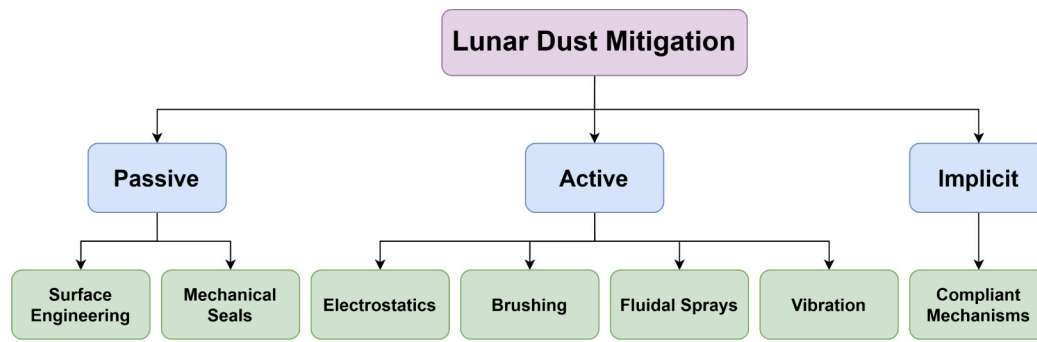


Fig. 4. Lunar Dust Mitigation Method Classification.

Table 1
Common Lunar Regolith Geotechnical Properties.

Property	Range	Units	Source
Bulk Density (ρ)	1.3–1.92	g cm^{-3}	Heiken et al. [29]
Porosity (n)	40–65	%	Connolly and Carrier [39]
Specific Gravity (G)	3.1	–	Heiken et al. [29]
Friction Angle (ϕ)	30–50	°	Heiken et al. [29]
Cohesion (c)	0.1–3.4	kPa	Connolly and Carrier [39]

The bulk density asymptotically approaches 1.92 g/cm^3 . Similarly, the intergranular porosity (the percentage of void space in a volume) of lunar regolith was found to vary from 65% at the surface to less than 40% at depth (asymptotically) [39]. The bulk density and porosity are inversely related; as porosity increases, bulk density decreases [29].

The shear strength of lunar soil can be described by the Mohr–Coulomb Failure Criterion, which breaks down the soil strength into cohesive and frictional components. Similar to density, the cohesion (interparticle attraction of same substance) and internal friction angle (a measure of shear resistance) of the lunar regolith increase with depth. Consequently, the shear strength of lunar soil rapidly increases after 15 cm, and again after 30 cm [39]. This means that larger excavation forces are required to excavate lunar regolith at greater depth and higher compaction. Many sources agree that regolith excavation at depths greater than 15–30 cm would be challenging, especially due to the low traction caused by the low-gravity environment [37–39]. This means that most excavation concepts focus on the extraction of surface regolith. The range of various geotechnical properties of lunar regolith is presented in Table 1.

Lunar regolith can also take on an electrostatic charge, and Van de Waals forces help dictate its movement [37]. The lunar regolith has low electrical conductivity and dielectric losses, preventing the regolith from dissipating accumulated charge, resulting in electrostatic charging under ultraviolet radiation. This phenomenon is particularly challenging for surface operations, since lunar regolith adheres electrostatically to whatever material it comes into contact with. The electrostatic charging effect is especially pronounced during the lunar night, when lunar regolith electrical conductivity decreases, causing more charge accumulation. The electrostatic charging effect can result in the coating of equipment, such as solar panels, potentially causing problems for lunar surface systems [29].

2.3. Lunar dust mitigation

Lunar dust has been documented by Apollo astronauts as being highly abrasive, highly adhesive, and electrostatically charged [29]. This can pose challenges for humans, but also machinery, since it can embed and compact itself into exposed cracks, cover and scratch surfaces, cut through materials, and damage electrical components by electrostatic discharge. This motivates the development of dust mitigation technologies, which can be broadly sorted into passive, active,

and implicit mitigation (or dust tolerant design), as shown in Fig. 4. Some recent reviews on lunar dust mitigation methods include the works of [45,46], as well as the trade study by [47]. Further, NASA has released a guide compiling their experiences and suggestions regarding lunar dust mitigation, including specific studies on optical surfaces, radiator and thermal control surfaces, communications and electronics, as well as mechanisms and mechanical assemblies [48].

Surface coatings and surface engineering, which aim to minimize the surface attraction of lunar dust, are the most common forms of passive dust mitigation. Of particular interest are hydrophobic coatings such as the *Lotus leaf coating*, a nano-textured coating composed of ceramic oxides developed by NASA, which has been tested on plastics, solar cells, metals, and ceramics. The coating leverages various nanoparticles of different sizes to reduce the surface energy, introducing hydrophobic properties such as water and dirt repulsion [49]. A slightly different method involves physical modification of a surface by laser ablation, creating surface topologies that naturally reject dust. In optimal conditions, this results in superhydrophobic surfaces [50]. Various etching methods have also been used to modify the surface topology of aluminum substrates, resulting in a four-fold reduction in lunar dust surface coverage [51]. Photolithography is used in another work for surface modification as well, with little success on polymeric materials and more success on silicon surfaces [52]. Alternatively, Polyhedral Oligomeric Silsesquioxanes (POSS) are organic solutions of macromolecules, which have been explored for topography modification, but concerns have been raised regarding their ability to withstand the harsh lunar conditions [53]. The important surface properties for dust mitigation have been investigated by treating and modifying the surface of Kapton, silicon, and quartz using an ion-beam process, finding that surface hardness and roughness are key factors to prevent the embedding of lunar dust in a material [54]. In general, these methods all aim to minimize the contact surface area or increase the surface hardness. Some less explored coatings include electrostatic discharge coatings and peel-away coatings, with more investigation needed to fully understand their potentials [46,47].

Seals are another form of passive mitigation, preventing dust from entering internal mechanisms. The spring-loaded Teflon seal has been demonstrated for dust mitigation on the Mars Exploration Rover (MER) Instrument Deployment device. However, questions remain regarding the seal's ability to withstand the extreme lunar temperature fluctuations and radiation conditions [55,56]. The Aeroflex rotary shaft seal [56] is a dry seal that has also been implemented on the MER. The drawbacks of this seal is that it requires high machining tolerances, can be misaligned when installed, and suffers under vibration. Lastly, there are magnetic coupling seals [47] that can protect rotary shafts, leveraging a magnetic field to transfer torque and create a hermetic seal between two sides. However, there are concerns that the magnetic field would attract the slightly charged lunar regolith.

Some works have sought to elevate the efficacy of passive methods by implementing active elements. For surfaces, one concept neutralizes the electrostatic charge of the lunar regolith, harnessing it for power

generation. As the electrostatic charge of the lunar regolith is neutralized, its adherence to surfaces is mitigated [57]. Another method seeks to actively alter the work function of a conducting material, effectively discharging the lunar dust [58]. For seals, work has been done to explore active mitigation methods for clearing dust that has already entered mechanical seals, achieving up to a 90% dust repulsion rate using a set of parallel screen and parallel plate electrodes [59].

In terms of active dust mitigation, many concepts leverage the electrostatic charge of the lunar dust. The most developed active dust mitigation technology is the Electrodynamic Dust Shield (EDS), which consists of a set of electrodes embedded in a dielectric film connected to an AC power source. The fields are turned on and off in an alternating sequence, moving the electrostatically charged dust particles across the surface [60,61]. Several versions have been developed at Kennedy Space Center, including glass integration and flexible fabric mesh (for spacesuits) [62]. The EDS has also been designed to clear dust from key habitat components, such as solar panels, radiators, optical systems, visors, seals, connectors, and doors [63]. Some recent research has sought to address scalability concerns, such as the Habitat Orientable and Modular Electrodynamic Shield (HOMES), which aims to provide self-cleaning modular building blocks for surfaces, floors, and door mats [64]. The EDS has been tested outside of the International Space Station (ISS) during the Materials on International Space Station Experiment (MISSE), yielding promising results [65–67]. A similar concept, the Electrostatic Lunar Dust Collector (ELDC), consists of a grid layer of parallel conducting plates aligned normally to the protected surface. These plates are connected to opposite ends of a DC power supply, inducing an electrostatic field that attracts oppositely charged lunar dust particles [68,69]. A 52.55% difference in collection efficiency is observed for triboelectrically charged dust particles in low versus high vacuum [70]. The Electrostatic Lunar Dust Repeller (ELDR) was developed to address concerns regarding the weight of plates, the partial blockage of solar radiation, and the cleaning requirements of the ELDC. The ELDR consists of a set of thin needle-shaped electrodes connected in parallel to one side of a DC power supply [71,72]. Since the lunar surface is generally slightly positively charged during the lunar day and negatively charged during the lunar night, the ELDR can be charged accordingly to repel lunar dust particles. A similar work implements carbon nanotube (CNT) technology to produce a dust trap/deflector, charging two parallel nanotube surfaces to attract dust passing through an opening [73].

To avoid lunar dust abrasion on sensitive surfaces, the Space Plasma Alleviation of Regolith Concentrations in the Lunar Environment (SPARCLE) operates using a series of electron or ion guns to charge lunar dust, then attracting it to the opposite cleaning electrode or wand for disposal [74]. Similar technologies have been developed using handheld magnetic devices for the removal of lunar dust from fabrics such as spacesuits. The first of these magnetic rollers leverages electrostatic waves to attract and trap lunar dust, with limited success [75]. The following generation of magnetic roller implements a series of neodymium magnets, taking advantage of the nanophase iron that can be found in lunar regolith [76]. A similar dust removal method leverages UV light radiation on a porcelain lanthanum modified lead zirconate titanate (PLZT) plate, creating high potentials between the plate and the surface to be cleaned [77,78]. Analysis has shown that greater illumination intensity leads to high voltage potentials on the dust removal plate, resulting in a faster response time [79]. Another surface cleaning technology is the electron beam, firing electrons that enter micro cavities by secondary emission, inducing a charge buildup in the non-conductive lunar dust particles. The charge buildup creates a repulsive force, clearing the surface over time [80]. Improvements are implemented by varying the beam incident angle [81] and using multiple electron beams [82]. It has also been proposed that dust collection or repulsion methods could be mounted on a lunar rover, such as an EDS repulsion system and a PLZT collection system mounted on the end of a lunar mobile manipulator [83].

Other studies have investigated the physical removal of lunar dust through brushing mechanisms. A NASA study on clearing thermal control surfaces tests numerous brushes, including nylon, PTFE, and Thunderon bristles, under laboratory and simulated lunar conditions. The brushes are tested on both AZ93 thermal control paint and an aluminized FEP (AlFEP) thermal control surface. The Escoda nylon fan brush and the Zephyr fiberglass fingerprint brush are effective on both surfaces, also successfully generalizing to simulated lunar conditions [84]. Brushes can also be improved with electrostatic dust mitigation methods, such as combining an electrostatic cleaning system with a negatively ionized Thunderon brush [85]. Similarly, TEST-RAD is a chinchilla-inspired brush, made from multifilament yarn, polyester, and Kapton, that integrates electrostatic mitigation, created by students as part of NASA's Big Idea challenge [86]. Within the same challenge, students also investigated the integration of electrostatic mitigation and UV light bombardment with a mechanical brush dust cleaning system [87]. These integrated approaches aim to minimize the accumulation of lunar dust on brush devices after use.

There exist some less explored unique lunar dust mitigation concepts. Cryogenic liquid sprays have been tested in vacuum on spacesuit-like fabrics, which is contrasted against gaseous nitrogen sprays in identical conditions, comparable to cleaning with compressed air in terrestrial environments [88,89]. Similarly, spraying high-pressure gas to dislodge dust particles has been considered, but little work has been done to further investigate the concept [47,48]. Another unique concept involves the vibration of surfaces at various frequencies to clear dust [90]. However, dust cleared by vibration may be prone to migrating to nearby surfaces. A compilation of the anticipated surface clearing efficiency for the various active lunar dust mitigation methods is presented in Table 2.

Finally, studies have explored the implicit mitigation of lunar dust in the design of various system components, focusing on the concept of compliant mechanisms. In general, components are often damaged by lunar dust when it enters gaps between adjacent elements. Compliant mechanisms seek to eliminate these inter-element gaps, reducing or eliminating the performance decline due to increased friction and jamming [58]. Furthermore, compliant mechanisms do not require lubricants, and are suitable for high-precision applications. Studies have demonstrated that some system components, particularly end-effectors, could be replaced with compliant mechanisms. A case study for the design of a geological tool (tongs) from the Apollo missions was undertaken, demonstrating the feasibility of the compliance concept for lunar dust mitigation [91,92].

3. Lunar infrastructure

The vast majority of proposed habitat locations on the lunar surface involve the lunar North and South poles. In 2022, NASA identified 13 candidate landing regions within 6 degrees of latitude of the lunar South Pole for the Artemis missions [16]. These missions would aim to establish a sustainable human presence on the Moon with the Artemis Base Camp [2]. NASA cites the presence of in-situ resources in PSRs, particularly water ice, as well as large periods of continuous sunlight, as some of the primary considerations for these selections [16]. Similarly, the ESA's Moon Village concept has identified areas around the Shackleton crater at the lunar South Pole as a target location [3,4]. The CNSA has also cited the lunar South Pole as a target habitat location, with some researchers conducting a thorough site selection based on the trade-off between scientific and engineering objectives, with a particular attention to the Amundsen and Malapert craters [17]. Another study investigates the lunar North and South poles as possible sites for a liquid mirror observatory, finding the North Pole to be optimal, due to the larger slopes at the Shackleton South Pole location, unless the habitat and observatory are decoupled [18]. Finally, a study for a *PneumoPlanet* inflatable habitat investigated possible habitat locations and landing sites within 1° angular distance from both poles, proposing ideal locations in both regions [19].

Table 2
Maximum Clearing Efficiency of Active Lunar Dust Mitigation Methods.

Method	Surface Clearing (%)	Source
Electrodynamic Dust Shield	>90%	[60–67]
Electrostatic Lunar Dust Collector	100%	[68–70]
Electrostatic Lunar Dust Repeller	100%	[71,72]
Space Plasma Alleviation of Regolith Concentrations in the Lunar Environment	Not quantified	[74]
Magnetic Rollers	64%	[75,76]
Porcelain Lanthanum Modified Lead Zirconate Titanate UV Light	99%	[77–79]
Electron Beam	92%	[80–82]
Brushing Mechanisms	66% - 99.5%	[84–87]
Cryogenic Liquid Sprays	98.4%	[88,89]
High-Pressure Gas Spray	60.44%	[47,48]
Vibration	>90%	[90]

Table 3
Comparison of Lunar Habitat Concepts.

Feature	Landed Self-contained Structures	Rigid Modules*	Inflatable and Deployable Structures
Structural Integrity	High	High	Moderate
Launch Volume	High	Low (AM)/High	Low
Launch Mass	High	Low (AM)/High	Low
Internal Volume-Launch Mass Ratio	Low	Low/High (AM)	High
Assembly Complexity	Low	High	High
Lifespan	Temporary outpost	Permanent base	Outpost or base
Radiation Protection	Low	Requires shielding/Moderate (AM)	Requires shielding
Pressurization	Pre-pressurized	Requires pressurization	Requires inflation
Transportability	High	Low	Moderate
Scalability	Low	Moderate/High	Moderate/High

* Rigid modules can be composed of either transported or in-situ (AM) materials, as discussed in Section 3.1.2.

3.1. Lunar habitats

Two recent reviews of lunar habitat structures and concepts are conducted by [12,93]. Some of the main types of habitat structures in the reviews include landed self-contained structures, rigid modules, inflatable and deployable structures, tensile-cable structures, tunneling or coring, exploited caverns or lava tubes. The strengths and weaknesses of the first three concepts are broken down in Table 3, and examples are further described in the following subsections.

3.1.1. Landed self-contained structures

Self-contained structures are popular, particularly for temporary habitation or initial landing. For example, the Mankins Habot is a module that would autonomously land on the Moon, and move away from its original landing location to allow for further Habots to land [93]. Once a set number of Habots land, they would couple to form a pressurized base. The Habots would be capable of decoupling and moving the entire base to another area of interest. It is suggested that a crew size of 4–8 would be ideal, with a minimum of 3 crew members needed to operate and maintain the units. More recent variations of the Habot concept have been proposed, leveraging technologies such as the All-Terrain Hex-Limbed Extra-Terrestrial Explorer (ATHLETE) to manipulate habitat modules on the lunar surface [94]. Some researchers

have also proposed combined rover-habitats, which are vehicles designed to navigate the lunar surface and provide temporary shelter for astronauts [95].

Another concept involves the use of the SpaceX Starship Human Landing System as the foundation for a lunar base [96]. The Starship would be placed horizontally on the lunar surface, and transformed into a habitable space on the lunar surface via a workforce of modular rovers and astronaut workers. The concept leverages the existing equipment on the Starship to run the habitat's operations. The major changes to the original concept include the lying down of the spacecraft, unpacking of permanent amenities, and burying of the spacecraft under 5 m of regolith. It is worth noting that the concept is dependent on the Modular Robotic Construction Autonomous System (MoRoCAS) [96] rovers that are capable of tasks such as drilling, moving regolith, and more; however, such a fleet of robots does not yet exist.

3.1.2. Rigid structures

Rigid structures allow for more permanent habitats on the lunar surface, with the best structural properties but the most challenging material mass requirements. One concept for a rigid structure habitat involves the use of at least 6 cylindrical modules launched by an ARIANE rocket with a payload of 12 tons [97]. This concept includes the Teleoperated Rocket Crane, which would be assembled in orbit to aid with the landing of the modules. The modules are made of

aluminum sheets with a double-shell structure, which is to be filled with lunar regolith to protect a crew of 8 astronauts from radiation, micrometeorites and temperature variations. Each cylinder is 17 m in length and 6 m in diameter (inner diameter of 4 m).

Another concept is an igloo-shaped magnesium structure supported by sintered regolith, and covered in sandbags for radiation shielding [98,99]. The thermal analysis of this structure shows that given a 3m-thick regolith shield, the structure can be thermally insulated at any location on the Moon. It is envisioned that such a structure would be pre-constructed by autonomous mini-robots prior to the arrival of astronauts. The proposed structure weighs 226 t and provides 470 m³ of working area.

One way to circumvent the high mass transportation requirements of rigid habitats is through ISRU, particularly leveraging Additive Manufacturing (AM). This is what the 3D printed Lunar Lantern habitat concept aims to do, through the collaboration of NASA and the private AM company ICON. The habitat is a three-story structure capable of supporting a crew of 4. The design process also considers environmental factors, such as lunar seismic activities and micrometeorite protection using 3D printed Whipple shield panels [100]. Similarly, the proposed Xuanwu lunar base concept aims to leverage AM to create sintered construction blocks, which will be robotically manipulated to construct the base. The base consists of a unique arch-based structure, with flexible solar panels being fitted to the top [101].

3.1.3. Inflatable and deployable structures

Inflatables have been extensively considered for the creation of a lunar base, due to their inherent advantages in total mass, volume, ease of deployment and more. One example consists of a large inflatable shell that is covered in a protective regolith layer [93,102]. The habitat is 16 m in diameter, and can accommodate a dozen astronauts. A relatively newer but similar concept is that of Moon Base Two [103], which aims to house 4–6 astronauts for 6 months. A more recent concept is that of Pneumoplanet, which proposes an inflatable habitat complete with 16 greenhouses to permanently house and feed up to 32 astronauts [19]. Some important concerns regarding inflatables include the sensitivity of the inflatable material to radiation exposure, the effect of the abrasive lunar regolith shield on the membrane material, and the behavior of the membrane under harsh temperature variations and vacuum [93]. Some recent concepts have addressed these concerns, such as the Clover Lunar Base concept, which aims to protect the inflatable membrane by 3D printing a protective shell. Similarly, the Red Star Lunar Base concept aims to protect the membrane using lunar regolith bag technology [104]. Another study also suggests the protection of an inflatable habitat membrane with lunar regolith bags, finding the concept to be mechanically feasible [105].

Hybrid structures are variations of inflatables, such as TransHab [93] or the Hybrid Lunar Inflatable habitat Structure (HLIS) [106], which have a solid core structure surrounded by an inflatable shell. Hybrid concepts are leading candidates for most current habitat concepts. For example, the ESA's Moon Village habitat is largely based on the TransHab morphology, with a height of 15.5 m and an inflated diameter of 10.5 m, supporting a crew of 4 astronauts [6]. Another hybrid concept is put forth by the Robotic Lunar Surface Operations 2 (RSLO2) project, combining an inflatable tunnel region (covered in regolith) with rigid habitation modules [107]. The RSLO2 habitat aims to support a crew of 4 astronauts for at least four 30-day stays per year. The RSLO project has been used to learn lessons regarding the autonomous deployment and operation of a potential Moon base. [108]. The current design for the initial Artemis Base Camp is also a hybrid concept, with a two-story inflatable section lying on top of a metallic base for egress and ingress. The habitat is 7.8 m tall, and has an inflated diameter of 6.5 m. The concept seeks to support a 2-person crew for 30 days and a 4-person crew during swap-outs [109,110]. A chronological review of inflatable and hybrid concepts for lunar habitats is conducted

by [103]. The hybrid concepts strive to achieve the low mass and volume benefits of inflatables, along with the structural advantages of rigid structures. Another similar variation involves inflatable rigidization, where an inflatable membrane is hardened by some chemical process to become a rigid structure. This would allow for easy initial transport and deployment, while benefiting from advantageous structural properties. However, rigidization may be challenging at the lunar South Pole, due to the extreme temperature fluctuations, emphasizing the importance of the glass transition temperature of the resins or coatings used [111].

Other concepts include non-inflatable deployables, such as Mobitat2 [112] made from Transformable Robotic Infrastructure-Generating Object Network (TRIGON) panels [113]. The TRIGON architecture uses equilateral triangular panels to deploy many different stable geometries. A recent work has demonstrated a self-deploying concept for the TRIGON architecture [114]. Another similar concept is a lunar habitat consisting of linear and planar elements connected through simple movable joints, which addresses the volumetric limitations of cargo mission flights [112]. The concept employs scissor-like elements (SLEs), a set of simple interconnected linear elements (SILE), a movable intermediate connection element (MICE), rigid panels, a spacing element, and a pressure barrier to build the structure of the habitat.

3.2. Lunar power stations

Various studies have investigated the challenges and possible technologies associated with the creation of a power microgrid on the lunar surface. One study investigates the power required for an oxygen production facility and base camp for 6 astronauts, finding an estimated daytime power of 172.5 kW, nighttime power of 28.05 kW, and peak power of 233.5 kW [115]. Another study finds a maximum power consumption of 23.11 kW for a base of 10 astronauts [116]. A study also considers the baseline NASA life support power consumption [117], as well as some lunar ISRU processes, finding an average power demand of 38.90 kW and a maximum power demand of 43.74 kW for 6 astronauts [118]. The RSLO2 study considers a few different lunar base locations and configurations for a crew of 4 astronauts, finding a power consumption of 1096 kW on the Shackleton slope using high voltage cables, versus a demand of 8209 kW deep inside the Shackleton crater using laser-beamed power transfer [107]. This variety of power consumption levels, ranging from 20 kW to over 1000 kW, shows a large dependency between the power demand and the base configuration, such as number of astronauts, base location, ISRU technologies, and power transfer systems. As such, it is likely that any power generation or storage system would need to be designed and sized according to the specific base configuration.

The two primary candidates for power generation are photovoltaic (solar) cells and nuclear fission reactors [115,119–121]. Hybrid systems, leveraging lunar power at some times (e.g., lunar day) and nuclear power at other times (e.g., lunar night), have also been investigated [119]. Although nuclear reactors have the advantage of being able to supply continuous power with minimal operational constraints, mitigating the need for a large Energy Storage System (ESS), they suffer from a lower power-to-mass ratio. The state-of-the-art solar cells have energy densities up to 350 W/kg [122], as compared to 5 W/kg provided by the latest space-rated nuclear reactors, e.g., NASA's KRUSTY kilowatt reactor [123]. Determining the ideal configuration for a lunar power station depends on many factors, such as the illumination conditions and the length of the lunar night. Many solar power concepts seek to leverage near-constant illumination conditions of a potential polar lunar base, finding that solar arrays gain a mass advantage over nuclear reactors at polar sites illuminated for over 75% of a lunar cycle [119]. This has led some researchers to propose power stations based solely on solar power, relying on the ESS to survive the lunar night [107,116,118]. However, for a solar array power system, as the length of the lunar night increases, so does the size of the ESS. It has been suggested that a hybrid solar and nuclear power system may offer

the best trade-off between mass and power production, with nuclear power taking over during lunar night [119]. It is also worth noting that even the most recent and advanced nuclear reactors designed for space have predicted power output of up to 10 kW, meaning that multiple reactors may be needed to power a single station. Further, the reactors would need to maintain a distance of at least 1.15 km from the inhabited base to keep radiation levels below the safe 0.05 Sv/year threshold [115].

Some novel power generation technologies have also been explored, but remain at low Technological Readiness Level (TRL), such as Fresnel lens-thermionic conversion [124], laser power transmission from solar power satellites [125], and power generation from electrostatically charged particles [126]. NASA has also been investigating numerous technologies for future power generation, including Low Energy Nuclear Reactions (cold fusion), Fusion, Metastable Nuclear Isomers, Positrons, Atomic Fuels, and Structural Bond Energy Release. These energy production strategies face significant challenges, such as the production of net positive energy for Fusion and Nuclear Isomers [121].

Lithium-ion (Li-ion) batteries and Regenerative Fuel Cells (RFCs) are the primary technologies considered for long duration energy storage on the lunar surface [115,119,120]. Although Li-ion batteries have the higher TRL, their specific energy level is typically around 200 Wh/kg [127], as compared to the 300 to 1000 Wh/kg provided by RFCs [115,128]. It has been noted that Li-ion batteries perform better for short discharge durations, while RFCs gain a mass advantage over the batteries for discharge durations over 10–18 h, such as the lunar night [120,129]. For a lunar base ESS, it has been estimated that RFCs would reduce the size of the ESS by a factor of three or four [119]. It has also been suggested that the PSRs of the lunar South Pole could be leveraged to store the RFC reactants at cryogenic temperatures, increasing the energy density of the overall system [130]. Another less developed technology is the Lithium-Sulfur (Li-S) battery, which can practically provide energy densities between 500 and 600 Wh/kg. However, current Li-S batteries are limited to 100 to 500 discharge cycles (100% discharge), as compared to 500 to over 1000 cycles for Li-ion counterparts [119,127]. Further, current Li-S batteries exhibit high self-discharge rates and high levels of capacity decay [131].

3.3. Lunar communications

Lunar communications will be a crucial element of building sustainable lunar habitats, particularly to support ISRU operations [132]. Researchers have established communications architecture as a “very high” priority factor during early missions until 2054 [133]. In general, lunar communications architecture can be broken into three categories: Earth ground stations, lunar surface communications, and relays and constellations. This section focuses on the lunar surface communication systems as well as relays and constellations. The Earth ground stations are typically multi-purpose for a range of interplanetary missions.

Several factors can impact lunar surface communications, including location and mission scenario, e.g., crewed vs. uncrewed. It is estimated that lunar surface communications to and from Earth as well as relay satellites will be done in either the Space Network (SN) S-band or the K/Ka band, whereas such communications for ExtraVehicular Activities (EVA) will take place solely in the SN S-band. Various communications scenarios require both low and high throughput bandwidth, with unmanned surface systems requiring 10–25 Mbps data rates and manned EVA systems requiring 50–100 Mbps [134].

One particularly challenging factor for lunar communications is the loss of line of sight to Earth, which occurs every 14 days at the South Pole [135]. To achieve consistent connectivity, it is generally accepted that a lunar communications and navigation network will need to be established. This network will be composed of Lunar Communications Terminals (LCTs), or *nodes*, and lunar relay satellites [134, 136]. Examples of proposed communications (and navigation) architectures include NASA’s LunaNet [136], ESA’s Moonlight [137,138], and

China’s Queqiao Integrated Constellation System [138,139]. To enable lunar surface communications architectures, there have been some endeavors to design lunar surface telecommunications networks and technologies to increase connectivity and network speeds. One concept investigates the usage of 5G networks for the lunar surface, using small lunar communications towers (15-m tall) and various surface nodes, i.e., base, rover and astronaut [140]. Nokia has done work to support the development of LunaNet through their lunar surface LTE communications system [141,142]. The Nokia system is set for validation efforts at the lunar South Pole aboard the Intuitive Machines Nova-C lander, equipped with a 4-m tall base station and a 1-m tall antenna on the lunar rover [138]. A number of initiatives have also investigated optical laser-based communication, which could provide higher data rates and power efficiency. These would involve optical receivers and laser transmitters that need to account for some additional factors, such as refraction through the Earth’s atmosphere [135,143]. The Australian National University (ANU) is currently developing an Optical Communications Ground Station (OCGS) to support laser-based communications [144].

The establishment of a lunar communications network also introduces the possibility of establishing lunar surface localization technologies, similar to the Earth’s Global Positioning System (GPS) or Global Navigation Satellite Systems (GNSS). Such technologies would likely be introduced through constellations, with researchers finding three spacecraft in halo orbits near the Lagrange points (L_1 and L_2) to be the minimum required to achieve 70% instantaneous coverage at any given time [145]. Other variations include the HyInc Walker orbits, achieving close to 93% coverage with 24 satellites in quadruple coverage performance [146], and a four-satellite constellation prioritizing polar orbits to provide radio-frequency-based navigation and communications [147]. Most constellations, including LunaNet and Moonlight, include Position, Navigation and Timing (PNT) as a part of the communications infrastructure [136–138].

3.4. Landing pads

A major challenge to the goal of repeatable lunar landings is the interaction of rocket engine exhaust plumes with the lunar regolith, which can create blast ejecta and cratering. The ejecta can reach velocities over 2000 m/s, and travel large distances in a vacuum, potentially entering orbit and cis-lunar space. It is estimated that the Apollo lunar lander displaced 2600 kg of regolith during landing, leading to erosion and cratering [148]. The ejecta creates high velocity particles that can damage surface structures, and remain suspended in orbit, and the cratering results in uneven terrain, both of which could hinder future lunar landings [149,150]. While current rocket and localization technologies may require landing pads of up to 100 m diameter, the landing area is projected to be reduced to 25 m as landing accuracy improves [100,151,152]. One trade study investigates the cost of candidate landing pad construction methods, dividing the task into an inner zone (12 m radius) and outer zone (27 m radius). The inner zone must withstand the high temperature and pressure of the plume stagnation region, while the outer zone must withstand erosion. The study broadly considers grading, compacting, sintering, paving, gravel, and polymer as candidates for the inner and outer zones [153]. Another trade study focuses largely on AM methods for the construction of lunar landing pads, considering only a few non-AM implementations [154]. Two recent reviews on lunar landing pads have been conducted by [149,150].

The Lunar Spaceport by the Open Lunar Foundation proposes four architectures for a potential lunar landing pad, culminating in a final robust landing pad with target construction in 2120 [155]. The first concept, The Skinny, represents the current state of technology, with an inner zone consisting of an octagonal platform made of metal panels, surrounded by an outer zone composed of a heat-resistant textile apron. Next, there is The Machine, which is a spacecraft-like system elevated

off of the ground with legs, and uses a robotic arm for servicing the visiting vehicle. There is also The Brute, which consists of a 3D printed high-temperature polymer pad on top of an elevated mound of regolith. The fourth concept is The Works, which would constitute a fully operational spaceport. This final concept would have a multi-zone pad, an ejecta shield, and subterranean systems for refueling and capturing exhaust gases. The pad itself would be made of grouted, oven-sintered basalt setts connected by liquid geopolymer.

A NASA study investigates the regolith stabilization problem for a vertical take-off and landing pad using ISRU, and finds that a combination of techniques would be ideal, such as combining solar sintering for an inner zone and polymers for an outer zone [156]. The study also conducts field testing using the PISCES rover and robotic arm, successfully demonstrating the fabrication and deployment of 100 sintered basalt interlocking pavers over a 3m×3m inner zone. The rover further constructs a 20m-diameter outer zone, leveling and grading the zone, then covering it in gravel [157]. Another recent concept for a lunar landing pad considers the compaction of surface regolith and the usage of stacked rocks for stabilization. The study finds that although simple gravel landing pads would not be effective, a multi-zone approach could be effective and feasible, where the central zone is made by interlocking pavers and the outer zone is made of gravel and rocks [158]. Another work also investigates the possibility of instant landing pads created using the Masten in-Flight Alumina Spray Technique (FAST), which would spray the surface upon arrival. The technique aims to mitigate the landing plume effects and to allow greater flexibility in landing location, particularly for scientific missions [159]. A different approach to landing pad construction is being taken by Project Olympus, a collaboration between NASA and the large-scale 3D printing company ICON, which seeks to leverage 3D printing technology to build whole habitats, including landing pads [160]. The technology has been tested on a subscale level in practice, exploring various morphologies for the landing pad structure [161]. Alternatively, it has also been suggested that lunar regolith bags could be used as a base for a lunar landing pad [149]. Overall, a robotic construction system for landing pads will realistically need to be able to accommodate a variety of construction techniques, catering to both the inner and outer zones.

3.5. Other infrastructure

Other infrastructure, such as blast berms and non-pressurized shelters, will also be necessary when establishing a base on the lunar surface [162]. With regard to using blast berms to protect surface structures during landing and take-off, simulations show that larger particles of the ejecta would be traveling with supersonic speeds (>343.2 m/s) at less than 3° from the touchdown, and smaller particles would be ejected with slower velocities at close to 17° [100,163]. A few different methods have been proposed for the creation of such blast berms, including piling raw regolith [164], additive manufacturing [100,155], and regolith sandbags [165,166]. One concept proposes the stacking of irregular and unprocessed lunar stones for the creation of blast berms [167]. It is also suggested that lunar stones could be stacked in three layers, with average rock diameters of 144.7 mm, 72.4 mm and 14.5 mm for the top, middle, and bottom layers, respectively [158].

Non-pressurized shelters may be used to protect equipment or surface systems. These could be constructed through deployable components [168], additive construction [100], or regolith sandbags [105]. Some studies have been undertaken to analyze the stability of potential non-pressurized regolith bag structures, finding regolith bag arch structures to be stable and feasible [105,169]. These structures could serve purposes such as robotic storage, server storage, or communications [166].

Although roads will also be a crucial piece of infrastructure for the expansion of future lunar bases, little work has been done to consider their construction specifically. A collaboration between NASA and The University of Texas at Austin investigates the construction of lunar

roadways and relevant technologies, including graders, compactors, blades, augers, and various drive configurations [170]. Another more recent study considers how the in-situ regolith could be used for the creation of lunar roadways [171]. The lack of emphasis on the research on lunar roadways can likely be attributed to the fact that the technologies used for building lunar landing pads can be similarly utilized for roadways, such as sintering and paving technologies, amongst others [172]. Most AM, sintering or paving technology research declares the creation of lunar roadways as a general goal, such as Project Olympus by ICON and NASA [100]. One specific lunar roadway building project is PAVER by ESA [173,174], which seeks to raise the TRL of casting and 3D printing technology for paving from 3 to 5. The project leverages a 12kW CO₂ laser with spot beam up to 100 mm to achieve the in-situ melting or sintering of the lunar regolith for the creation of roadways and other paving elements.

4. Lunar construction methods and materials

There exist a few reviews of current research on lunar construction materials, including a survey by [13] and a review of NASA's developments by [14], as well as a trade study by [175]. A broader review has also been conducted recently, examining the work done in extraterrestrial construction methods and materials, including those for the lunar environment [176]. These methods and materials are used for the construction of larger infrastructure elements, e.g., habitat shells. Internal elements, such as piping and electronics, would likely need to be pre-constructed, and are not discussed in this review.

4.1. Regolith binding

In general, the use of regolith-based conventional concrete suffers from its water consumption requirement, resulting in the need for either transporting large quantities of water to the Moon or sourcing water through ISRU. Other challenges include temperature variations and vacuum, which would prevent the use of water for tasks such as curing [13]. The first proposed waterless concrete is sulfur-based concrete [177], which enhances durability under certain exposure conditions, and accelerates hardening, achieving 70-80% strength within 24 h [178]. One study reports producing sulfur concrete with up to a maximum of 47 MPa compressive strength [179]. Despite these characteristics, it is noted that sulfur concrete could deform and melt under lunar temperatures that could rise up to 128°C , which is higher than the melting point of sulfur [180]. Researchers have also turned to polymer-based concretes, where polymers are used as the binder material. One particular concrete uses a 2-8% epoxy binder, and can produce concrete with comparable or superior flexural strength than typical steel reinforced concrete samples [181]. Studies have reported maximum flexural strengths for polymer-based concretes ranging from 16 MPa [182] to as high as 60 MPa [181]. It has also been found that epoxy-based concrete is a viable, fast and simple method for creating lunar concrete. Another type of concrete is geopolymer-based, using inorganic polymers made by chemically activating aluminosilicate materials with an alkaline solution. Although geopolymers require water in their activator solutions, the final structure does not include water [183]. Some works report very high compressive strengths for such concretes, while other reports show poor results under lunar simulated conditions. For example, one study in particular finds up to 19 MPa average compressive strength for geopolymer concretes cured in ambient conditions, dropping to an average of approximately 3 MPa when cured in simulated lunar conditions [184]. Such uncertainty concerning the curing process in the lunar environment remains the largest barrier for all binder methods, since it is difficult to determine how strong the materials cured in those conditions will become. However, it has been suggested that since the gravitational force on the Moon is about $1/6$ g, the structural requirements for cement would also be $1/6$ of the terrestrial requirements [184]. Although building codes vary by region,

the U.S. Army Corps of Engineers standards suggest that concrete structures should have a compressive strength of at least 25–30 MPa [185], which can then be estimated at 4–5 MPa on the lunar surface. Another challenge facing binder methods is the increased up-mass required to transport binders from Earth. For example, a 10% epoxy binder for 1 tonne of building material requires 100 kg of mass transported from Earth.

4.2. Additive manufacturing

Additive Manufacturing (AM) has become an increasingly popular concept for lunar construction. In some cases, binder materials are used in the AM process. One ESA-funded project leverages D-shape 3D printing technology, mixing granular simulant with pulverized metal oxide, then spraying it with a liquid binder and adding a second layer on top. These layers are then compressed, successfully creating various building blocks with 20.35 MPa compressive strength and 7.11 MPa flexural strength [186]. However, what makes AM attractive is the possibility of using minimally pre-processed regolith through sintering methods. One study implemented the direct-extrusion of JSC-1 A lunar regolith simulant mixed in an ink with poly(lactic-co-glycolic acid) copolymer and organic diluents to create micro-trusses. The materials are then sintered, achieving between 1 and 19 MPa compressive strength, depending on sintering duration [187]. A more recent study applies compaction during the sintering process to produce samples with enhanced compressive strengths, reaching up to 67.68 MPa [188]. Another form of sintering is laser sintering, where high energy lasers are used to heat the lunar regolith. This method is shown to have high geometrical accuracy [189], with one study achieving a 22.7 MPa compressive strength on laser-sintered samples [190]. One newer concept employs Laser Powder Bed Fusion, a method which uses a high brilliance laser beam to melt successive layers of powder. Although this method is slower than D-shape, it can have improved mechanical properties, with one experiment achieving compressive strengths up to 31.4 MPa [191], and another achieving up to 50 MPa [192]. Microwave sintering is also an option, with one study using a 2.45 GHz microwave furnace to sinter lunar regolith simulant, achieving compressive strengths up to 37 MPa [193]. Direct sintering of lunar regolith blocks has also been done successfully, with one study reaching compressive stress values of 218.8 MPa at 1.44% porosity and 84.6 MPa at 11.78% porosity [194].

A major challenge facing AM technologies, particularly sintering-only technologies, is the extremely high input energy requirement [195], which becomes a particularly important problem on the Moon, especially for large habitat-scale projects. To minimize the input energy requirement, studies have investigated the implementation of solar sintering technologies, leveraging the often constant illumination conditions on the lunar surface in some regions. One experiment demonstrates the creation of solar-sintered lunar regolith blocks using JSC-1 A and JSC-2 A simulant, achieving 5 MPa compressive strength [196]. Another experiment produces a compressive strength of 2.49 MPa for solar-sintered samples in ambient conditions, but has limited success in a vacuum [197]. Solar concentrators have also been utilized to perform fusion welding of some metal parts, such as carbon steel and titanium, using lunar regolith simulant as a binding agent, achieving tensile strengths of 2.94 MPa (for carbon steel) and 4.95 MPa (for titanium) [198].

4.3. Raw materials

Almost all habitation concepts for a lunar base involve the utilization of regolith for shielding from the harsh lunar environment [93]. As such, it becomes necessary to investigate the best way to move and shape the regolith. Beyond movement of the raw regolith, some works have investigated the possibility of containment units, typically of “sandbag” form, as a construction unit for various structures. It has been argued that such Regolith Containment Units (RCUs) benefit

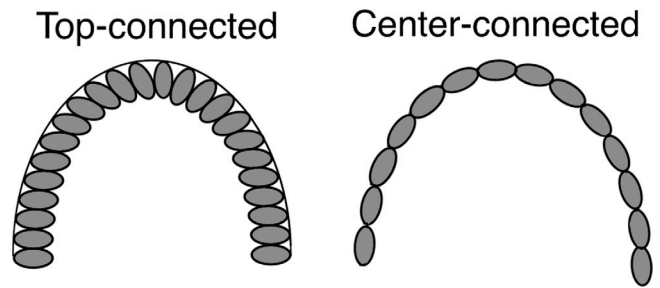


Fig. 5. Top-connected versus Center-connected RCUs.

from a high in-situ material ratio, short production time, and less stringent temperature requirements, suffering only from slightly lower compressive strengths as compared to competing methods [175]. An early study on such lunar regolith containers is conducted by NASA, investigating potential materials and filling mechanisms. A container filling mechanism is designed, but not tested, using a conveyor and funnel system with a theoretical capacity of filling 120 container per hour for 8 h a day. The containers are selected to be 0.0283 m³, and made of a fiberglass composite weave [199]. A similar study suggests a scoop mechanism for filling 0.487 m³ containers made of Kevlar 149. With such a mechanism, it is estimated that the system could fill 10 containers per hour for 8 h a day [200]. Another conceptualization on utilizing RCUs for building protective structures, such as berms and shelters, is discussed in [201].

One particular NASA project involves the creation of a lunar RCU to build a garage-type structure [202]. A multitude of materials are considered and tested, including Vectran, Zylon, Twaron, Nextel, Nomex, and Gore PTFE. After testing, Vectran is recommended for further study, followed by Kevlar and Twaron. The project also outlines various approaches to filling the containers, including shovels, funnels, tubes, sandblasters, leafblowers, etc. In the end, the best approach is determined to be a hopper attached to a helical flexible screw conveyor system built by Hapman Inc. The final structures are erected at Marshall Spaceflight Center (MSFC). The work is continued by investigating the stability of these structures [169]. It is found that structures made from top connected containers are more stable as compared to center connected containers, both of which are constructed using frames (examples shown in Fig. 5). It is also found that a screw conveyor system is the most reliable method for filling the containers. Another study also considers potential container filling mechanisms, focusing on use cases for the pneumatic lunar regolith excavator from Honeybee Robotics [203]. The work proposes a few conceptual operational configurations for the excavator, including filling containers directly at the mining site before manipulation, centralized filling at a lunar regolith silo, and filling the containers at their final location. However, these concepts are not demonstrated beyond the proof of concept only for the pneumatic excavator. Another study investigates the design of an RCU filling and stacking system for the burial of a lunar outpost [204]. The design involves a wall-scaling system whereby the regolith is taken in, the RCU is formed and ejected directly on top of the stack. The final concept is tested using sand on a one-tenth scale.

A recent concept proposes containers equipped with microprocessors, piezoelectric power generators, RF transponders, and RFID tags for potential implementation in a lunar environment [165]. The work argues that RCUs present the lowest entry cost barrier to building lunar infrastructure, since they do not require the extraction of in-situ resources or the modification of regolith as much as some other technologies do, e.g., AM. The concept involves a collection of such containers to form a structure, which are able to actively communicate health and placement data to the assembly and monitoring systems. The work continues to consider the high-level design of the surrounding

architecture, such as the lunar regolith excavation, the RCU manipulation and filling subsystems [166]. The study focuses on the creation of berms and superadobe structures for equipment protection and storage, as well as protection of greenhouses and habitats. Another recent study also proposes aramid fabric RCUs as key components in habitat construction, testing their capabilities as load bearing columns, self-supporting arches, and external protection for habitat structures. It finds that RCUs provide a viable construction material when used in tandem with pre-fabricated and inflatable structures, with a column compressive strength of 6.9 MPa and 90% strength retention after vacuum aging [105].

Although the Moon's surface has been mostly pulverized, rocks and gravel still exist in different regions [205]. The use of raw materials in the form of lunar rocks and gravel is also investigated, particularly for the construction of landing pads [158,206]. These studies find that, although the central area of a lunar landing pad would need to be solid, the outer blast shield could be composed of loose gravel and rocks of varying sizes organized into three size-distributed layers. The required rock sizes are found to range from 14.5 mm, for the bottom layer, to 144.7 mm, for the top layer. It is calculated that the volume of rock required for the landing pad increases with the squared radius, resulting in 3140 m³ of rock needed for a landing pad of 100 m in diameter.

4.4. Analysis of methods and materials

The methods and materials for lunar surface construction have been a heavily researched field since the Apollo missions in the early 1970s. A taxonomy of various lunar construction methods and materials is suggested by [175], which is used in Table 4 to survey the related works in the literature. These works focus primarily on studies that have produced or analyzed sample materials.

Fig. 6 shows a percentage breakdown of the work conducted in the development of lunar construction methods and materials. From the figure, nearly equal efforts are made for Reaction and Bonding Solidification categories. However, Sintering/Melting has dominated the field, which can be attributed to a larger variation of methods, with the most research being done on direct sintering (9.3%), microwave sintering (7.2%), and laser melting (7.2%). Figs. 7 and 8 depict the time evolution of lunar construction methods and materials categories and subcategories, respectively. The research on sulfur cement (13.4%) dominates the Bonding Solidification methods, largely due to its precedence in the field. Yet, it has been recently succeeded by polymer cement (8.2%) and geopolymer cement (14.4%), which seek to address the shortcomings of sulfur cement (e.g., low melting point). The Confinement Formation category, so far merely consisting of regolith bagging (8.2%), originated in the 1990s, has recently experienced a resurgence with concepts proposing regolith sandbags for habitat protection. The Sintering/Melting category has particularly dominated over the past fifteen years, with many recent concepts finding new energy-efficient ways to heat the lunar regolith.

5. Lunar construction systems

5.1. Lunar regolith excavators

A taxonomy for classifying lunar regolith excavation systems is provided by [297], where they are primarily divided into discrete versus continuous systems. Discrete excavators need to break contact with the soil in between cuts to clear the cutting surface or dump excavated materials, e.g., front-end loaders, dozers, backhoes, scrapers, scoopers, etc. Continuous excavators have multiple cutting surfaces that are continually in contact with the soil, e.g., bucket wheel, bucket drum, pneumatic, bucket ladder, etc. These two classes are further broken down into complete versus partial systems, where the former includes a mobility system and the latter does not. The taxonomy can

be used to further expand the timeline that was originally presented by [288] for NASA's robotic excavators, as shown in Table 5.

The most recent NASA development is the one by Kennedy Space Center (KSC) Swamp Works, which is a series of continuous-complete dual-bucket drum excavators. They focus on excavating the top 5 cm of surface regolith by implementing a dual-counter rotating bucket drum system, which mitigates the effects of low gravity conditions. The bucket drums, designed by Lockheed Martin, are actuated by single Degree-of-Freedom (DoF) arms. The systems are tested with both sand and BP-1 lunar regolith simulant, in regular and icy conditions, and with gravity off-loading [298]. There have been three iterations of such excavation systems, namely, Regolith Advanced Surface Systems Operations Robot (RASSOR) [290], RASSOR 2.0 [299], and ISRU Pilot Excavator (IPEX) [291]. The first prototype, RASSOR, is teleoperated, but RASSOR 2.0 features some autonomous modes, and was able to handle a regolith payload of 80 kg with a system mass of 66 kg (down 50% from RASSOR). In terms of operation, RASSOR 2.0 expends 4 W of power per kg of regolith excavated, and is capable of excavating a minimum of 2.7 metric tons of regolith per day. A recent series of papers have described various aspects of the newest iteration, IPEX, including bucket drum scaling [300], wheel testing [301], and the design of the system to Technological Readiness Level (TRL) of 5 [291], all with the aim of preparing for a future demonstration mission. The IPEX is approximately 0.7 times the size of RASSOR 2.0, with a regolith payload capacity of 30 kg.

The Lunar Excavation and Size Separation System (LES³), mounted aboard the lightweight (40-60kg) LUVMI-X lunar rover [292], is a discrete and complete lunar regolith excavator prototype. The mechanism has a mass of less than 2 kg, and is capable of scooping 100 g per charge, while maintaining excavation forces below 8 N. The excavation is done by implementing a cylindrical scooping mechanism, specifically designed for its low excavation forces. The concept focuses on excavating only the surface layers of lunar regolith to avoid the compaction of lunar regolith at depth. The LES³ uses a combination of dry sieving and vibration to separate lunar regolith into two size categories for storage. The vibration also serves to reduce the necessary excavation forces.

Another high-level concept of excavation and long-distance conveyance system is a combined bucket chain excavator and ballistic conveying system, involving "shooting" the regolith container over a distance to the destination [302]. Initial Discrete Element Method (DEM) modeling for the design of the bucket chain excavator system has been performed. Although low gravitational forces and hard vacuum make this method possible, the deployment logistics of such large and complex mechanisms without prior infrastructure have not been considered yet. Another study investigates a bucket-ladder system for the excavation of lunar regolith, built by the Planetary Surface Technology Development Lab (PSTDL) at Michigan State University (MTU), aiming to raise the its TRL to 5 or 6 [294]. Previous experiments have shown that bucket ladders can have excavation rates greater than 1000 kg/hr at 0.01 W/kg/hr in atmospheric conditions, but the work aims to extend the previous experiments to more realistic lunar environments. A major downside of bucket-ladder systems is the higher risk of mechanical failure, due to a higher number of moving parts. The study proposes the evaluation of wear and tear of bucket-ladder systems in the lunar environment as a major objective, but little results are presented.

Researchers have also developed CASPER, a novel continuous and complete screw-propelled excavation rover for lunar or Martian use [293]. The excavator is based on the Archimedes screw propulsion system concept, leveraging the screw turning direction to push regolith into the storage system. The optimal configuration for the system achieves a 30kg/h excavation rate with system mass of 3.4 kg, a power draw of 30 W, and a speed of 10.2 m/min while excavating. It is important to note that silica sand was used as an analog for lunar regolith. To produce higher fidelity results, the system should be

Table 4
Lunar Construction Methods and Materials Summary.

Ctg.	Method	Description	Citations
Reaction Solidification	Dry-Mix/Steam-Injection (DMSI)	Cures Portland cement mixes in a saturated steam pressurized environment.	[207–212]
	Sorel Cement	A magnesium oxychloride cement.	[186,213]
	Geopolymer Cement	Uses lunar silicoaluminate materials and minimal water consumption.	[183,184,214–225]
	Hydration	Traditional cement cured using water-based techniques.	[226,227]
Bonding Solidification	Polymer Cement	Mixes polymeric binders with lunar regolith.	[181,182,228–233]
	Sulfur Cement	Involves melting sulfur and mixing it with lunar regolith.	[177–180,234–242]
	Biomaterial Cement	Involves the usage of bio-based materials as binders.	[243–246]
	Metal Bonding	Uses metallic powders or solutions mixed with lunar regolith.	[247]
	Cold Sintering	Applies pressure, a NaOH solution, and minimal heat to sinter lunar regolith.	[248]
Sintering/Melting	Casting	Complete melting of lunar regolith into a molten state.	[249–252]
	Laser Melting	Uses high-powered lasers to melt regolith powders layer by layer.	[174,189–192,253,254]
	Solar Melting	Concentrated solar energy is used to melt lunar regolith.	[196–198,255]
	Digital Light Processing	Photocurable resin forming solidified through sintering.	[256–259]
	Self-propagating Sintering	Uses heat from exothermic reactions (Al/Mg) to sinter regolith.	[259–265]
	Microwave Sintering	Sintering via microwave dielectric heating.	[193,266–271]
	Direct Sintering	Electrically heating lunar regolith, typically binder-free.	[187,188,194,241,272–276]
Conf. Form.	Regolith Bagging	Involves packing lunar regolith into bags.	[105,165,166,169,199, 200,202,203]

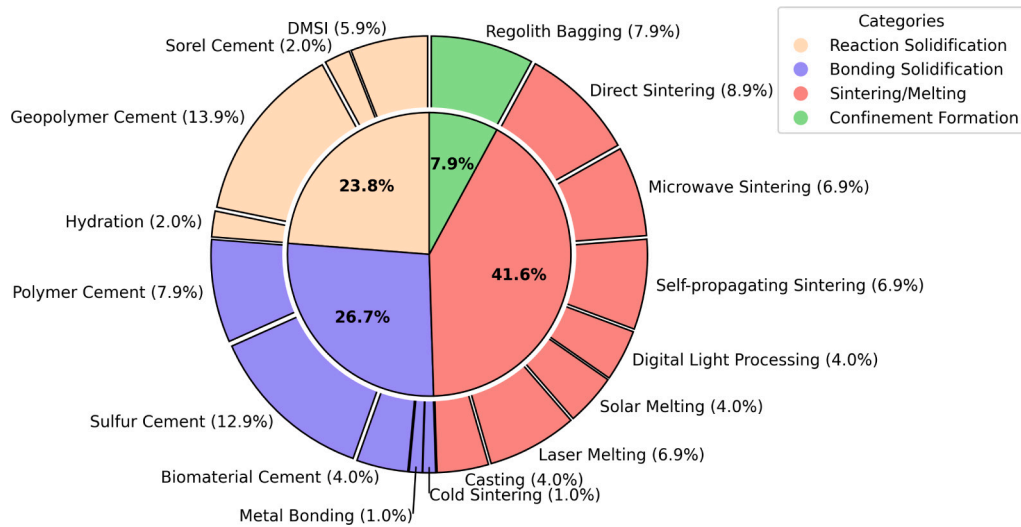


Fig. 6. Lunar Construction Methods and Materials Categories.

tested on lunar regolith simulants, which can have drastically different properties than typical sand. Some similar pre-cursor concepts have been tested, largely focusing on the auger or screw mechanism within the context of drilling and flexible screws [303], or regolith conveyance and extraction of surface regolith [304].

The developments around NASA's lunar excavators have been promoted by a number of initiatives. The concept studies, performed by

Eagle Engineering Inc. in 1988, were a part of the NASA Space Exploration Initiative (SEI) efforts to begin developing lunar construction technologies, including excavators, such as bucket-wheel excavators, front-end loaders, and scrapers, among others [305]. The development of lunar excavators was continued through the NASA Regolith Excavation Competition (REC) from 2007 to 2009, involving teams of university students, with a total of 49 excavator prototypes being tested

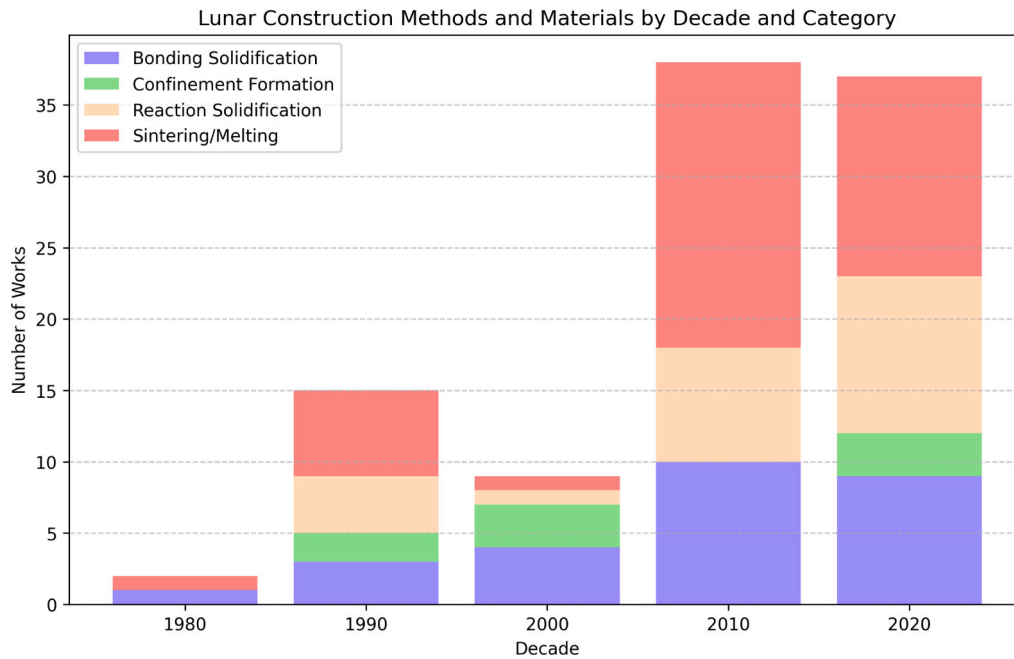


Fig. 7. Lunar Construction Methods and Materials Development Over Time.

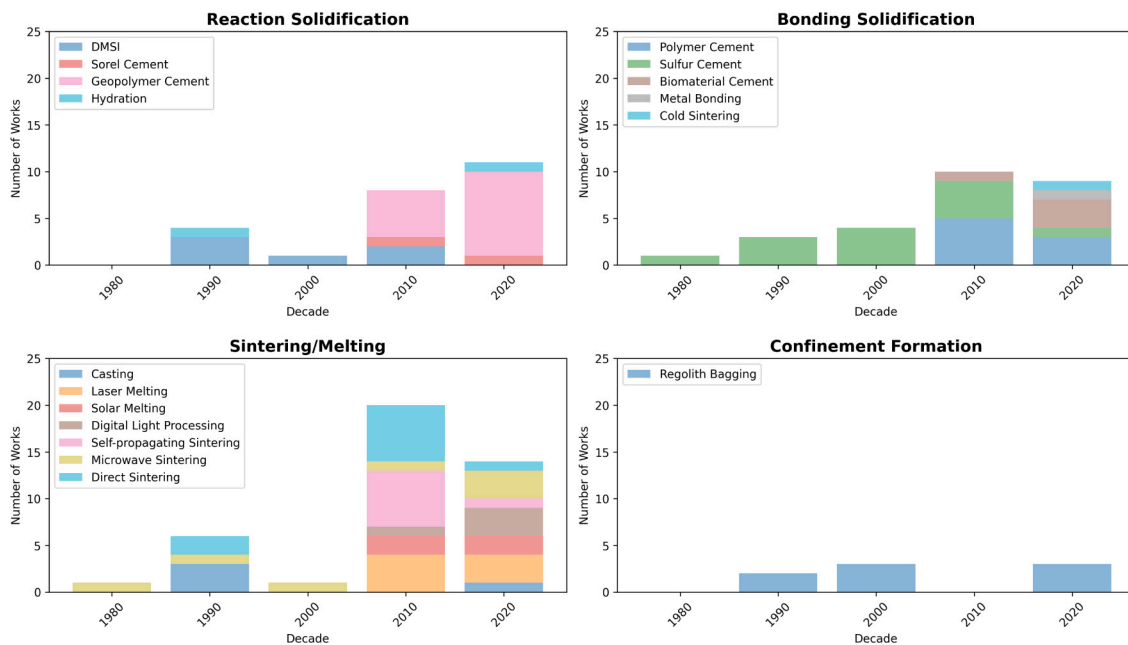


Fig. 8. Lunar Construction Methods and Materials Sub-Categories Over Time.

with JSC-1a lunar regolith simulant [298]. An important initiative towards the advancement of lunar excavators has been the Desert Research and Technology Studies (RATS) field analog tests [306]. Many of the excavators and mobility systems developed by NASA (and collaborators such as the CSA) in the late 1990s until the early 2010s were tested under this initiative. Since 2010, NASA has been hosting the “Lunabotics” Robotic Mining Competition for teams of university students, leading to over 500 excavation prototypes within a decade [307]. In this competition, a testbed is filled with BP-1 lunar regolith simulant that is expected to be excavated in a short period of time, e.g., 15 min. Throughout the competition, the most popular excavation concepts have been bucket-ladders and front-end loaders. Recent iterations of the Competition have also required the excavation of a minimum 1 kg of icy regolith simulant [37].

5.2. Lunar regolith conveyance

Similarly to the excavation techniques, lunar regolith conveyance methods can also be broken down into three categories, namely, batch, continuous, and transfer. Batch methods involve the discrete conveyance of finite amounts of lunar regolith using systems such as haulers, aerial cableways, railcars, ballistics, sleds, and robotic arms. Continuous methods keep the lunar regolith in constant motion using systems such as drag chains, conveyor belts, vibrating surfaces, augers, bucket-ladders, and pneumatic conveyors. Transfer mechanisms allow lunar regolith to be transferred between different points in the system, including at intake. Some examples are auger-and-hopper mechanisms, hoppers, and bowl feeders. A review and assessment of lunar regolith conveyance methods is conducted by [47], examining long-distance

Table 5
Lunar Excavator Prototype Timeline.

Developer	Period	Name	Excavator	Type*	Source
Colorado School of Mines	2001–2011	Bucket Wheel Excavator	Bucket Wheel	CC	Muff et al. [277]
NASA	2007	Cratos	Scraper	DC	Caruso et al. [278]
Lockheed Martin	2008–2010	Bucket Drums	Bucket Drum	CP	Mueller and Van Susante [279]
CSA	2008–2012	Juno Rover with Load-Haul-Dump	Front Loader	DC	Sanders and Larson [280]
Carnegie Mellon	2008–2016	Scarab	Bucket Wheel	CC	Skonieczny et al. [281]
Honeybee Robotics	2008–2019	PlanetVac	Pneumatic	CP	Zacny et al. [282]
NASA	2009–2010	Chariot with LANCE	Dozer Blade	DC	Sanders and Larson [280]
SysRand Inc.	2009–2010	Moonraker	Bucket Chain	CP	Gao et al. [283]
NASA	2009–2013	ATHLETE with VIPER	Backhoe	DC	Mueller et al. [284]
NASA	2010–2012	Space Exploration Vehicle with LANCE	Dozer Blade	DC	Mueller et al. [285]
Astrobotic inc.	2010–2016	Polaris	Bucket Wheel	CC	Skonieczny et al. [286]
NASA	2012–2016	Cantaur with Glenn Digger	Front Loader	CD	Bauman et al. [287]
NASA	2013–2021	Centaur with APEX and Badger	Front Loader	CD	Mueller [288], Proctor et al. [289]
NASA	2010–2019	RASSOR	Bucket Drum	CC	Mueller et al. [290]
NASA	2019–2024	IPEX	Bucket Drum	CC	Schuler et al. [291]
University of Manchester	2021	LUVMI-X with LES ³	Scoop	DC	Just et al. [292]
Arizona State University	2021	CASPER	Screw	CC	Green et al. [293]
Michigan State University	2023	Bucket Ladder Excavator	Bucket Ladder	DP	Guadagno et al. [294]
South Dakota State University	2023	Lunar Bucket Wheel Excavator	Bucket Wheel	CP	Lohsandt [295]
Huazhong University	2024	Bucket Drum Excavator	Bucket Drum	CP	Li et al. [296]

*Discrete-Partial (DP), Discrete-Complete (DC), Continuous-Partial (CP), Continuous-Complete (CC)

conveyance, short near-horizontal conveyance, short near-vertical conveyance, and transfer points. Wheeled haulers are recommended for long distances, while conveyor belts, vibratory systems, and augers all perform well for shorter distances, and robotic arms are also viable options if regolith is pelletized. Auger-and-hopper systems with gravity discharge are found to be the best for transfer points along a conveyance route, with robotic arms also performing well for pelletized regolith.

Despite the many student prototypes related to the Lunabotics excavation competition, little work has been done to formally raise the TRL of many conveyance concepts. Some recent works have investigated vibratory conveyance, with a few studies testing the concept in both vacuum and atmospheric conditions. One study uses two dielectric elastomer actuators to vibrate an acrylic plate with lunar regolith simulant, enabling a maximum mass flow rate of 1.95 g/s in Earth gravity. It is found that 30 Hz vibration frequency at 60° inclination angle would result in optimal conveyance [308]. Another study investigates the conveyance of lunar regolith simulant by the vibration of a small (26-mm diameter) tube using a piezoelectric actuator, instead of vibrating a flat surface, achieving a mass flow rate of 7 g/s. The study also finds that regolith can be vibrated upwards at an angle of up to 24° in Earth conditions [309]. A follow-up study uses a similar actuator setup, but incorporates some numerical calculations, through DEM and single-particle modeling, to predict the conveyance performance in lunar gravity. Considering gravity and air drag, calculations predict that the conveyance system would be 1.4 times more effective in lunar conditions, achieving conveyance speeds of nearly 25 mm/s [310]. A more recent work tests an eccentric mass vibratory conveyor, where a motor spins two eccentric masses while connected to a pulley. The system achieves an average lunar regolith simulant mass flow rate of 209.86 g/s at a frequency of 3.46 Hz in Earth gravity and in vacuum [311].

A number of experiments have been conducted by NASA Kennedy Space Center (KSC) to develop and test a pneumatic conveyor for feeding an ISRU oxygen production reactor, in both 1g and $\frac{1}{6}$ g gravity [312, 313]. Both air and helium are used for the transport of various lunar regolith simulants, achieving mass flow rates of up to 6.4 kg/min [314]. As a part of the same series of experiments, augers are also field-tested at NASA's Mauna Kea Analog Site for feeding lunar regolith to the oxygen production reactor [315–317]. Flexible screw conveyors have also been conceptualized and tested, enabling the curved transport of lunar regolith [303]. One particular auger system is found to be capable of conveying an average of 298.35 g/s of lunar regolith simulant in atmospheric conditions, and an average of 215.71 g/s in vacuum [311].

Hoppers are another promising conveyance technology, with much work being done to investigate the optimal outlet widths and inclination angles in both 1g and $\frac{1}{6}$ g conditions. One experiment achieves a maximum mass flow rate of 22 g/s in $\frac{1}{6}$ g using a 75° inclination and 18 mm outlet width [318]. Another experiment achieves a mass flow rate of over 15000 g/min with a 5cm funnel opening in Earth conditions [319]. However, such experiments further conclude that it remains difficult to reliably predict the flow characteristics of lunar regolith.

Several experiments have also been conducted on other conveyance methods. A concept leverages the electrostatic nature of the lunar regolith, using electrostatic traveling waves, to transport regolith at a rate of 13.5 g/min [320]. Another conveyance technology also involves the electrostatic or magnetic transport of lunar regolith particles. One experiment successfully demonstrates the conveyance of lunar regolith using one or multiple solenoid coils along a tube with 200 mm length and 16 mm diameter at a 45° upwards angle [321]. The work then extrapolates the results, and concludes that this method could propagate 1.54 g/pulse of lunar regolith over a 1m long tube. A more recent conveyor belt concept is the Tandem Of Lunar REgolith CONveyors (TOLRECON), a rod scraper conveyor that horizontally transports lunar regolith to a bucket elevator system. The concept has been simulated and field-tested as a part of the Over the Dust Moon Challenge by the Colorado School of Mines [322].

5.3. Lunar robotic construction systems

The existing concepts for lunar robotic construction systems and technologies are primarily divided into four groups: cranes, mobile manipulators, 3D printers, and robot teams (see Fig. 9). This section is focused on works that are targeting the design of potential flight systems or have undertaken prototyping to this end.

5.3.1. Cranes

A number of generations of lunar cranes have been conceptualized and demonstrated (see [15] for a survey.) Early lunar crane concepts include the Lunar Excursion Vehicle Payload Unloader (LEVPU) by NASA [323] and the Stewart Platform Instrumented Drive Environmental Robot (SPIDER) by the National Institute of Standards and Technology [324]. The most recent and advanced examples of lunar cranes are the multiple generations of NASA's Lunar Surface Manipulation System (LSMS), with the first generation being proposed and

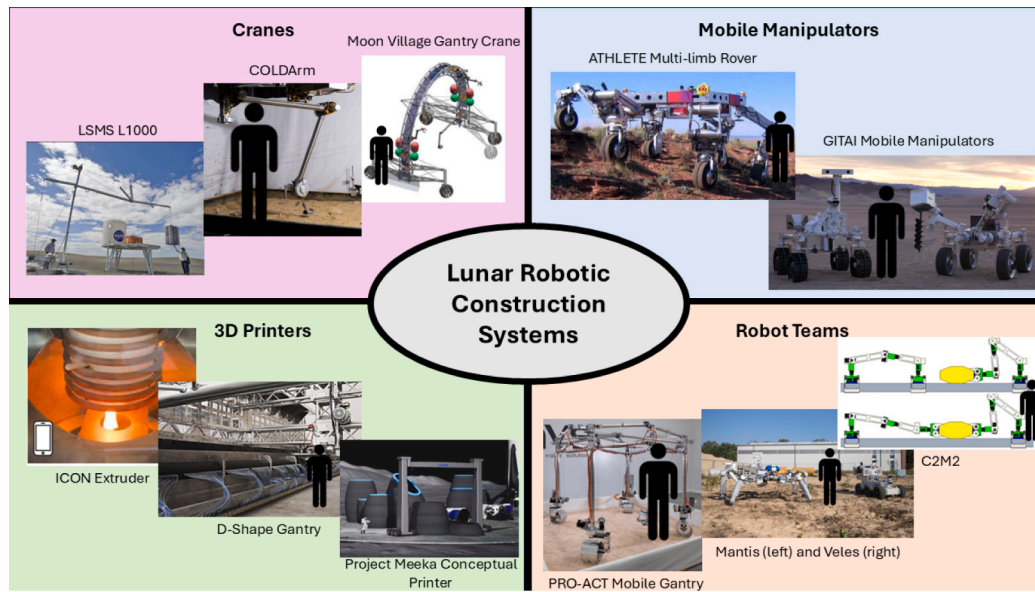


Fig. 9. Overview of Lunar Robotic Construction Systems.

Table 6
LSMS Variations.

LSMS Generation	Lander	Height (m)	Reach (m)	Lunar Wrist Capacity (kg)	Mass (kg)
Generation I	Altair	3.75	7.5	1000	309
Generation II	Altair	4.25	8.5	3000	190
Single-Link LSMS	Altair	3.75	7.5	4879	108
LSMS-H (Heavy)	Altair	4.17	8.34	6907	169
L1000	Blue Moon	4.0	8	1000	100-150
L500	Griffin	–	5.0	500-600	–
L35	Peregrine	0.9	2.25	35	7

field-tested in 2008 [325–327]. The cable-driven system is proposed for payloads too heavy for astronauts to handle, ranging from 100 kg to 3000 kg. The mass-to-payload ratio for the first generation is 3%, emphasizing its ability to manipulate large loads at low mass. The first generation is later enhanced with automated control [328] and hardware improvements [329,330]. The second LSMS generation has also been designed in parallel, with a focus on reducing the weight of the system and implementing self-offloading capabilities. The second generation also includes a single-link variation that would largely act as a crew elevator to and from the lander, including the lifting of incapacitated astronauts, along with payload offloading. The work has been continued through the design of a heavy-lift variant, the LSMS-H [331], which focuses on the offloading of more massive cargo, such as a habitat module. The LSMS-H also explores the possibility of mounting the LSMS onboard the Chariot mobility base to enable operations with a third-party ESA lander, with a focus on determining the tip-over boundary. The development of the LSMS has been continued, with the newest generations reaching the protoflight stage, including the L1000 [332], L500, and L35 [333], with 1000 kg, 500-600 kg, and 35 kg load capacity, respectively. Notably, the LSMS L35 has been selected for a technology demonstration as a part of the LSMS AutoN-omy capabilities Development for surface Operations and construction (LANDO) project, being paired with Astrobotic's Peregrine lander for payload offloading. Further developments on the LSMS continues, with recent studies exploring the reproducibility of the LSMS [334], safe operational envelope [335], and trajectory tracking [336]. A summary of the various LSMS generations is presented in Table 6.

As part of the ESA's Moon Village project, two mobile crane configurations have been conceptualized for transporting the habitat from the landing site, a gantry type crane and a pick-and-carry type crane. The gantry crane concept is based on the mobile gantry system for the RSLO2 project (Section 3.1.3), which builds off of the ATHLETE mobility platform (Section 3.1.1), achieving a 15000 kg carrying capacity

with only a 6600kg mass. Each gantry can be equipped with up to four robotic arms and a variety of gripper, maintenance, and excavation tools. It has been proposed that one lander can deliver up to two gantry systems, each deploying their mobility system upon landing [107].

The Cold Operable Lunar Deployable Arm (COLDArm) is a manipulator that can function as a crane, designed by NASA and intended to be mounted on the deck of a lunar lander [337]. The arm has 4 DoFs, and is approximately two meters long, enabling a tip payload of ~65 kg on the surface of Moon. The arm has been designed to work in extremely cold cryogenic temperatures without the use of heaters, enabling its operation in extreme environments, such as the lunar night. Eliminating heaters helps reduce the input energy required and heat exposure during volatile collection. The COLDArm has been tested with a variety of lunar regolith simulants, demonstrating its ability to operate in the lunar environment.

RoboCrane [338] is an alternative usage of lunar crane technology, being used to support power and communication architectures. Specifically, RoboCrane aims to assist with the exploration of lunar lava tubes and caves. The system would lower a rover of approximately 400 kg into these tubes or caves and support the subsequent exploration (or other tasks) by providing a power and communication interface with the surface.

5.3.2. Mobile manipulators

The development of mobile manipulators for lunar construction has recently entered the prototyping phase. The Pacific International Space Center for Exploration Systems (PISCES) has created a mobile manipulator named Helelani [157]. The four-wheeled rover has been field-tested in lunar analog conditions in Hawaii, as a part of the vertical takeoff vertical landing (VTVL) pad experiment. The rover was used as a robotic manipulator for paving the landing site, as well as a leveler equipped with a leveling back-blade. Another wheeled manipulator

concept is the Chinese Super Mason (CSM), with a 6-Degree-of-Freedom (DoF) arm for construction on the Moon [101]. The manipulation concept is implemented using the ABB IRB 6700 6-DoF manipulator. The robot arm is demonstrated in a controlled Earth environment, building a $2.8\text{ m} \times 0.8\text{ m} \times 1.1\text{ m}$ arched-framework structure in 8.5 h. However, a fully integrated concept including the mobility platform for the arm has yet to be demonstrated. The Lunar Exploration and Robotic System (LERS) is a similar robotic platform [339], featuring a 6-DoF manipulator mounted on a 4-wheeled base. It is envisioned that the system could support a variety of lunar surface activities, including construction and infrastructure development. The design considers lunar environmental factors, such as terrain, and a preliminary prototype has been assembled, but testing has yet to be performed in either lunar analog or terrestrial environments.

The ATHLETE wheel-on-limb mobility platform has also been demonstrated and conceptualized with various interchangeable manipulators and tools to support lunar construction (Section 3.1.1). In its typical configuration, the ATHLETE mobility platform has 6 multi-link robotic limbs attached to a flat (interchangeable) base. The minimal configuration, called the Tri-ATHLETE, is composed of 3 limbs. Each limb is capable of functioning independently as a 6- or 7-DoF manipulator with interchangeable end-effectors. These end-effectors are often wheels to enable mobility configurations, particularly for high-capacity transport, but can also include other implements for tasks such as grappling, drilling, construction, additive manufacturing, and many more. The ATHLETE limbs can also operate independently, with the ability to function as independent manipulators or in unique configurations, such as a high-capacity crane [340]. The ATHLETE individual limbs have lunar payload capacities ranging from 500 kg to 11456 kg, while the complete mobility system capacity ranges from 1000 kg to 22912 kg. Overall, the design of the ATHLETE system has focused on prioritizing interchangeability in operation, allowing it to be modified to tackle a diverse range of tasks.

More recently, a four-wheeled rover base capable of mounting various manipulators is being developed by GITAI [341]. Two of these manipulators are the S1 and Inchworm robot manipulators. The S1 manipulator has 9-DoFs (including the hand), a maximum reach of 1 m, and it is capable of enacting 100 Nm of torque [342]. The Inchworm manipulator emphasizes the development of multi-purpose tools, with the ability to be transferred between the rover, lander, habitat, or other modules. It is a 7-DoF manipulator, plus two DoFs for the end-effector, with a 2m reach, capable of enacting 368 Nm of torque [343]. The GITAI mobile manipulator has demonstrated its ability to perform a variety of tasks, such as resource excavation, welding, and solar panel construction. The system has also been field-tested on the ISS and in lunar analog settings, and has demonstrated features mitigating lunar environmental effects such as lunar dust [344].

5.3.3. 3D printers

The investigation into contour crafting technology for lunar construction began in the early 2000s at NASA MSFC [345], investigating a variety of feed materials, including traditional wet-mix cement, DMSI, and sulfur/waterless concrete. The initiative also investigated various extrusion technologies, starting with a double-roller extruder, an auger extruder, and a vibration valve extruder, before developing a novel propeller-based extrusion system. The extrusion has been tested on a KUKA 6-DoF industrial robot manipulator to develop multi-layer walls and half-domes, with authors stating that a robot manipulator is preferred over a gantry-type system to eliminate size restrictions (i.e., structure does not need to fit within gantry workspace). The mixer and extrusion system was found to be stable and durable after over 500 h of testing. The same initiative also investigated Selective Separation Sintering (SSS), which is a powder-based AM method that can build parts out of metals, ceramics, and composites. A prototype sintering system was developed and tested, creating ceramic tiles,

bronze half-cones, and other objects. The results of the study are summarized extensively in a 2017 technology report [346].

Some researchers have been developing Mobile Robotic Arm for 3D-printing (MIRA3D), a mobile manipulator equipped with a Powder-Feed Fused Deposition Modeling (PF-FDM) printhead for processing lunar regolith [347]. The complete system has been simulated [348], prototyped, and further developed to equip the MOONRISE payload [349]. The MOONRISE payload demonstrates the selective laser melting (SLM) of lunar regolith simulant, creating small (3g) structures. Another technology for melting lunar regolith involves solar concentrators, demonstrated by the Regolight project [350]. The project includes the demonstration of a TRL-4 mobile printing head, capable of sintering regolith simulant, as well as a regolith feeding system and the conversion of CAD models into printing paths. Some studies have specifically investigated the design of other AM printheads for lunar regolith, such as the study at NASA KSC to produce a Zero Launch Mass (ZLM) 3D printhead [351]. The ZLM printhead has been demonstrated using regolith simulants, producing a 1-meter diameter ogive structure. Another molten regolith extrusion printhead is under development at the DLR, successfully demonstrated using lunar regolith simulant under vacuum conditions [352]. All of these printhead and extrusion systems will require actuation to move across the working area. This motivates the need for novel specialized printing systems, such as larger cable-driven printers, the development of which has begun in recent years [139,353].

Other 3D printing technologies are also under development through commercial ventures, such as D-Shape [354]. Full-scale structures have already successfully been created using D-shape, and the technology is being adapted to the lunar environment through usage of regolith simulants as feedstock and testing under vacuum. A company called D-Shape has since been established to continue developing the technology for both terrestrial and space applications [355]. In another initiative, NASA has entered a collaboration with ICON to develop Project Olympus [100], a conceptual additive manufacturing system for the construction of a lunar habitat using in-situ materials by 3D printing. The concept also includes the design of a 3D-printed habitat, named the “Lunar Lantern” for its protective outer shield structure. The collaboration aims to build roads, berms, garages, landing pads, and vertical structures. To support these objectives, ICON is developing the technologies required for additive manufacturing on the lunar surface, including lunar test facilities, lunar regolith simulants, and molten regolith extruders, as well as space-grade robotic manipulators and actuators [160]. Similarly, Project Meeka by the Australian company Luyten aims to extend their terrestrial building-scale 3D printing technology to the Moon through the microwave sintering of lunar regolith; however, not much information regarding the project has been publicly released [356]. A survey of 3D printing concepts, challenges, and methods can be found in [357]. Overall, although much research has been conducted within the field of AM for lunar construction, it is still a relatively new field with a few existing advanced prototypes for 3D printing on the lunar surface.

5.3.4. Robot teams

The MultiRob 3D project is an early collaboration on lunar robot teams between the Institute of Space Systems and the Institute of Aircraft Design in Germany [358]. The objective of the project is to design a multi-robot system for the AM of components on the lunar surface, including an excavator, transport rover, service robot, and lunar AM factory [359]. Some work has been conducted to test the AM technology using lunar regolith simulant [360]. A different study has also investigated multi-robot teams for AM [361], designing a series of prototypes to this end and testing them in a sand-filled testbed. Although the objective of the study is to produce a habitat regolith shield, it stops short of producing additively manufactured components.

More recent studies have investigated collaborative manipulation for lunar surface operations. As mentioned in Section 3, tall lunar

towers may be essential pieces of power generation and communication infrastructure. To maintain and assemble parts of these towers, the scaling Climbing Collaborative Mobile Manipulators (C2M2) are proposed [362]. Although the final concept envisions teams of such robots working collaboratively to support the towers, only a single 6-DoF double-ended manipulator has been considered so far. Another study investigates the autonomous assembly of solar panels using dual robot arms [363]. While the work focuses on the development of autonomous control capabilities, its prototype has demonstrated some success in the solar panel assembly task.

The European Space Agency has been developing PRO-ACT, a heterogeneous team of robots working towards the common tasks of surveying, construction, assembly, and transportation [364]. The agents in the PRO-ACT team are IBIS, a 6-wheeled 6-DoF mobile manipulator, and MANTIS, a multi-legged robot with 6 extremities and two 7-DoF manipulator arms. While IBIS is a mobile manipulator designed for pyrotechnic operations and reconnaissance, MANTIS is designed mainly for mobile manipulation. The IBIS robot uses its six independently driven wheels to navigate difficult terrain at speeds of up to 10 km/h. It possesses an extendable arm with more than 3 m reach. The MANTIS robot has a height of more than 1.7 m and weighs 110 kg, using its multiple extremities to navigate difficult terrain. Its 61 actively controlled DoFs allow it to navigate complex scenarios and manipulation tasks. The newest iteration of PRO-ACT features Veles, the next generation of IBIS [365]. The Veles robot includes various hardware improvements, as well as a 7-DoF manipulator capable of manipulating up to a 20 kg payload. More recently, the Veles robot has been tested to perform a variety of construction tasks, including grading, manipulation, and tool exchange [366]. A mobile gantry system, similar to that produced by RSLO2, has also been designed for the PRO-ACT team [367]. The gantry system has a workspace of 1.7 m $\times$ 1.7 m $\times$ 1.7 m, a lifting capacity of 500 N, and a system mass of 230 kg. Much of the PRO-ACT project has also focused on developing multi-robot autonomy capabilities, which has been a limiting factor towards the implementation of robot teams.

6. Conclusions and prospects

Spacefaring nations around the globe have shown an increasing interest over the past two decades in establishing permanent lunar infrastructure, pointing to the Moon as the next frontier in the expansion of humanity's sustained presence. While the current efforts are primarily driven by exploration, the endeavor will soon pivot to more commercial activities, such as accessing resources both on the Moon and throughout our Solar System. While the Moon provides several advantages over Earth for expanding space activities, such as low gravity, availability of local propellants, and reduced impact to Earth ecosystems, it is an exceedingly inhospitable environment. The paper presented the breadth and depth of current developments on the lunar infrastructure as well as construction methods, materials and systems that are instrumental to establishing a sustained human presence on the lunar surface, after providing an overview of lunar environment and its challenges along with lunar site surveying for a permanent base.

Despite the vast literature on various aspects of the lunar surface environment, e.g., temperature variations, high dose radiation, constant micrometeorite bombardment, extreme lighting conditions, etc., the properties and behavior of the lunar regolith and dust still remain a key area of research, due to the limited availability of real samples. The lunar regolith, with its unique characteristics, would be the most valuable in-situ resource for lunar construction, both in its raw and processed forms. The lunar dust, on the other hand, is commonly considered as the most challenging environmental factor that surface systems will need to manage, leading to much work in the field of lunar dust mitigation, with systems such as the Electrodynamic Dust Shield achieving flight readiness. With respect to the site surveying, the lunar South Pole has been identified as the primary target for most potential base sites for its unique properties, such as the possibility of

near-constant solar illumination at peaks of light and the presence of volatiles in permanently shadowed regions.

Various concepts of lunar infrastructure have been proposed since the Apollo missions, with recent habitat concepts leveraging new technologies, such as building-scale additive manufacturing. For permanent habitats at the lunar South Pole, sustained power stations are paramount, for which solar power generation and regenerative fuel cells appear to be the promising power generation and storage technologies, respectively. Beyond habitat infrastructure, lunar landing pads are of particular importance to enable easy and repeatable access to the lunar surface. Although there exist some proposed concepts for these landing pads, just a few complete systems have been field-tested so far. Further, current rocket localization technologies require 100m-diameter landing pads for safe landing, which could be reduced to 25 m as landing accuracy improves. The advancement of rocket localization systems should be a priority to enable lunar surface development, since it would drastically reduce the size of the required landing pads, thus the resources required to build the landing pads and blast berms.

The research on methods and materials for establishing the required lunar infrastructure is actively ongoing, with a particular emphasis on the in-situ resource utilization technologies to minimize up-mass. Despite a notable advancement in both regolith binding methods and additive manufacturing, the uncertainty in the composition of the lunar regolith at the South Pole and how the lunar surface conditions will affect material properties would necessitate proof-of-concept demonstration missions on the lunar surface. The technologies required to produce the materials in situ, particularly for additive manufacturing, are currently being developed through commercial partnerships, such as the one between NASA and ICON. Similarly, the idea of using raw materials as feedstock appears promising for the protection of surface structures from projectiles, thermal variations, and radiation, but the robotic systems required to create structures from the raw materials (regolith or rocks) have yet to be fully designed and field-tested for the lunar environment.

Much work has already been done to produce lunar robotic excavation, regolith conveyance and construction systems, albeit the need for more flight-ready systems. Lunar excavation systems have been in developments since the early 2000s, with much testing being done through student competitions and analog field tests, e.g., D-RATS. The IPEx system, developed by NASA, is currently the only flight-ready lunar excavation system, planned to be tested on the lunar surface in the coming years. Lunar regolith conveyance systems have been tested using simulants in a variety of emulated lunar conditions, with much progress being achieved in vibratory, auger and hopper conveyance. Studies of lunar regolith excavation and conveyance have highlighted the challenging nature of the regolith, including high shear strength, abrasiveness and adhesion, among others. As for lunar robotic construction systems, the proposed primary morphologies include cranes, 3D printers, mobile manipulators, and robot teams. Lunar cranes have undergone much development through NASA's Lunar Surface Manipulation System, which is ready for flight demonstration. In contrast, although early experiments are promising, the technologies required for 3D printing on the lunar surface, such as regolith extruders, are still in the development phase. Regarding mobile manipulators and robot teams, some remarkable advancements have been achieved, including those developed by GITAI, performing a variety of tasks such as excavation, welding and solar panel construction. Taking into account the challenging lunar environment in the design process, particularly the lunar dust and regolith, will be key to the success of any lunar robotic construction system.

Over the past few years, the international lunar surface missions have been learning more about the surface environment, including sample return missions, as well as performing technology demonstrations for a variety of commercial landers [368]. These include China's Chang'e 5 and 6 sample return missions, with the latter being the first sample return mission from the far side of the Moon. India's

Chandrayaan-3 became the first spacecraft to achieve a soft landing at the lunar South Pole, and Japan's Smart Lander for Investigating Moon (SLIM) achieved a soft landing at the Moon's equatorial region, becoming the fourth and fifth nations to achieve a soft landing on the Moon. In the private sector, Intuitive Machines and Firefly Aerospace have both landed commercial lunar landers, paving the way for future surface initiatives.

The NASA Artemis roadmap aims to establish initial lunar surface infrastructure by the late 2020s. As lunar surface missions transition from exploration to construction, it becomes important to consider some of the logistical challenges, such as building practices and project management. Some experts have begun to define building requirements for the lunar surface, such as bearing capacity estimation [369]. The beginnings of a lunar building code is proposed and outlined by [370]. Yet, much more work is needed to further develop a complete code. As lunar surface construction begins, practical considerations of how lunar construction projects will be managed will become crucial. Generally, extra-terrestrial construction projects can take inspiration from terrestrial construction, finding parallels in the tasks required to build structures [371]. The Lunar Surface Construction & Assembly Equipment Study considers the necessary processes and actions to build and assemble various structures on the lunar surface, but does not go as far as establishing a project management system [305]. A more recent work draws analogies between lunar base construction and terrestrial megaprojects to help establish a management structure and project implementation plan [372]. Investigations of this sort have continued to determine the necessary architectural elements and phases for the construction of a lunar base [373,374], but still lack specificity in the management processes. Such practical considerations will be integral to the success of lunar surface construction projects.

As more technologies move from conceptual stages to proof-of-concept prototyping, it is becoming increasingly important to design and build terrestrial lunar analog facilities. Such large-scale facilities are imperative to raising the TRL of lunar construction technologies, with a handful already in existence, such as the Lunar Analog Facility in Europe [375], the NASA Ames Lunar Testbed [376], the Granular Mechanics and Regolith Operations (GMRO) Lab at NASA KSC [377], and the Dirty Thermal Vacuum Chamber (DTVC) at the Korean Institute of Civil Engineering and Building Technology (KICT) [378]. Various small-scale analog environments have also been created, such as the system presented in [379] as well as those designed for specific tasks including drilling [380,381] and excavation [382]. These testbeds, nevertheless, lack the generalizability and scale required for ground-testing lunar construction technologies. Works have also been initiated to formalize the development process of such lunar regolith analog facilities and testbeds [383], which could simplify the development of these platforms in the future.

CRediT authorship contribution statement

Ketan Vasudeva: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M. Reza Emami:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Cameron Dickinson:** Writing – review & editing, Funding acquisition.

Declaration of competing interest

We hereby confirm that the authors have no conflict of interest with respect to the submitted manuscript.

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Data availability

No data was used for the research described in the article.

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